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**Adaptive STRIDE-based Multi-Label
Intrusion Detection using
AI-Augmented Behavioural Modeling**

by

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Abstract

Existing Intrusion Detection Systems (IDS) treat each network flow as belonging to a single threat category, discarding the structural reality that real attacks simultaneously violate multiple security properties. A backdoor intrusion, for instance, combines Spoofing, Repudiation, and Elevation of Privilege in a single event yet conventional classifiers reduce this to one label and move on.

This thesis reformulates intrusion detection as a structured multi-label prediction problem aligned with the STRIDE threat framework (Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege). Instead of $f(\mathbf{x}) \rightarrow$ single label, we build $f(\mathbf{x}) \rightarrow [S, T, R, I, D, E]$, assigning each network flow an independent probability score across all six threat dimensions simultaneously.

To support this reformulation, we introduce eight STRIDE-aligned behavioural feature signals including a Traffic Asymmetry Ratio, Session Entropy Index, and Stealth Probe Index that encode domain security knowledge directly into the feature space. These signals are validated through ablation study and SHAP-based explainability analysis, which independently confirm that each signal captures the security-relevant patterns it was designed to target.

The detection pipeline combines a static XGBoost ensemble (six parallel binary classifiers) with a temporal Bidirectional LSTM with soft-attention mechanism, fused via three ensemble strategies and refined through per-label threshold optimisation. The complete framework is evaluated on two independent benchmark datasets UNSW-NB15 and CIC-IDS-2018 without any dataset-specific modification between evaluations.

On UNSW-NB15, the framework achieves Macro AUROC of 0.9556 ± 0.0006 across five-fold cross-validation and Macro F1 of 0.6869 a gain of +0.2761 over single-label baselines. McNemar’s statistical significance test confirms that improvements are statistically significant across all six STRIDE dimensions ($p < 0.0001$). On CIC-IDS-2018, the identical pipeline achieves Macro AUROC of 0.9893, improving over UNSW-NB15 performance on all four metrics and ruling out dataset memorisation. SHAP analysis independently discovers `sttl` as a universal cross-threat indicator and `feat_asymmetry_ratio` as the primary predictor of Repudiation both consistent with established network security knowledge.

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Abbreviations

AUROC	Area Under the Receiver Operating Characteristic Curve
BE	Bachelor of Engineering
BiLSTM	Bidirectional Long Short-Term Memory
CIC	Canadian Institute for Cybersecurity
CV	Cross-Validation
DNN	Deep Neural Network
DoS	Denial of Service
EoP	Elevation of Privilege
FTP	File Transfer Protocol
GPU	Graphics Processing Unit
HL	Hamming Loss
HTTP	Hypertext Transfer Protocol
IDS	Intrusion Detection System
IoT	Internet of Things
IP	Internet Protocol
LSTM	Long Short-Term Memory
mAP	Mean Average Precision
ML	Machine Learning
MLP	Multi-Layer Perceptron
NIST	National Institute of Standards and Technology
OWASP	Open Web Application Security Project
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
SHAP	SHapley Additive exPlanations
SIEM	Security Information and Event Management
SMOTE	Synthetic Minority Over-sampling Technique
SOC	Security Operations Centre
STRIDE	Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege
SVM	Support Vector Machine
TCP	Transmission Control Protocol
TTL	Time-To-Live
UDP	User Datagram Protocol
UNSW	University of New South Wales
XGBoost	Extreme Gradient Boosting

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Chapter 1

Introduction

Every day, organizations face thousands of network intrusion attempts. Some are obvious brute force attacks hammering login portals. Others are subtle a backdoor quietly established over weeks, blending into normal traffic. Detecting these threats automatically is what Intrusion Detection Systems (IDS) are built for. But here is the uncomfortable truth: most IDS still answer only one question "is this traffic an attack, or not?" and even the more sophisticated ones at best name a single attack type per network flow. This matters more than it might seem. Real attackers rarely do just one thing. A backdoor intrusion, for instance, involves pretending to be a legitimate user (Spoofing), hiding evidence of activity (Repudiation), and gaining unauthorised access (Elevation of Privilege) all simultaneously. An IDS that labels this simply as "Backdoor" and moves on has captured one dimension of a six-dimensional problem. Security analysts working from such outputs are, in effect, making decisions with incomplete information. The problem is well documented in the literature. Moustafa and Slay introduced the UNSW-NB15 dataset specifically to address the limitations of earlier benchmarks, noting that real modern attacks exhibit complex, multi-faceted behaviours that single-label classifiers struggle to represent [1]. Subsequent work applying XGBoost [2], deep neural networks [3], and LSTM-based models [4] to UNSW-NB15 has produced strong binary and multi-class results yet all of these approaches retain the same fundamental constraint: one label per flow. Meanwhile, the STRIDE threat modeling framework

— Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege has been a cornerstone of structured security engineering since its introduction by Microsoft [5]. STRIDE gives security teams a shared vocabulary for reasoning about what properties of a system are under attack at any given moment. It is widely used in software design, risk assessment, and penetration testing. What it has never been used for, until now, is as a machine learning prediction target inside an IDS pipeline. This paper closes that gap. We reformulate intrusion detection as a structured multi-label prediction problem: instead of $f(x) \rightarrow \text{single label}$, we build $f(x) \rightarrow [S, T, R, I, D, E]$, where each network flow receives an independent probability score across all six STRIDE threat dimensions simultaneously. A backdoor attack no longer gets one label. It gets a threat profile high Spoofing, high Repudiation, high Elevation of Privilege that directly informs the response a security team should take. To achieve this, we combine a static XGBoost ensemble with a temporal Bidirectional LSTM model enhanced with an attention mechanism, capturing both per-flow feature patterns and sequential attack progression across time windows. We introduce eight behavioural feature signals including a Resource Exhaustion Index, Stealth Probe Index, and Asymmetry Ratio specifically designed to align with STRIDE threat semantics. We validate the framework using SHAP-based explainability [6] to confirm that the model is learning genuine security-relevant patterns rather than statistical noise. Critically, we test the entire pipeline on two independent benchmark datasets UNSW-NB15 [1] and CIC-IDS-2018 [7] and apply McNemar’s statistical significance test [8] to confirm that improvements over single-label baselines are real and repeatable. The results are strong. On UNSW-NB15, our framework achieves a Macro AUROC of 0.9556 ± 0.0006 across five-fold cross-validation. On CIC-IDS-2018, without any modification to the pipeline, performance improves further to 0.9893 confirming that the framework generalises across datasets. McNemar’s test returns $p < 0.0001$ across all six STRIDE dimensions, with an overall Macro F1 improvement of +0.2761 over single-label baselines. The main contributions of this paper are as follows:

We are the first to systematically reformulate intrusion detection as structured multi-label STRIDE vector prediction, enabling simultaneous scoring of overlapping threat

dimensions per network flow. We introduce eight STRIDE-aligned behavioural feature signals that encode domain security knowledge directly into the feature space, validated through SHAP analysis. We propose a hybrid temporal fusion architecture combining static XGBoost with Bidirectional LSTM and attention mechanisms, refined via per-label threshold optimisation. We demonstrate statistically significant improvement over single-label baselines across all six STRIDE dimensions ($p < 0.0001$, McNemar’s test), with an overall Macro F1 gain of $+0.2761$. We validate cross-dataset generalisation on CIC-IDS-2018, achieving Macro AUROC of 0.9893 surpassing UNSW-NB15 performance without any pipeline modification.

The remainder of this paper is structured as follows. Chapter 2 reviews related work. Chapter 3 presents our behavioural feature engineering and datasets. Chapter 4 describes the model architecture. Chapter 5 reports experimental results. Chapter 6 presents the ablation study isolating the contribution of each pipeline component. Chapter 7 covers statistical validation. Chapter 8 presents SHAP explainability. Chapter 9 discusses limitations, and Chapter 10 concludes.

Chapter 2

Related Work

2.1 The Evolution of Intrusion Detection Systems

Intrusion detection is not a new problem. Researchers have been working on it since the late 1980s, when Dorothy Denning first formalised the concept of anomaly-based detection [Den87]. Since then, the field has gone through four distinct phases — each one responding to the limitations of the one before it.

The earliest IDS relied entirely on signatures. Systems like Snort [Roe99] and early commercial IDS products were built around databases of known bad patterns, flagging anything that matched. They worked reasonably well against known threats. The problem is obvious in hindsight they are blind to anything new. A novel attack pattern that has never been seen before simply walks straight through. Signature-based systems also require constant manual updates, which makes them expensive to maintain at scale.

The response to this was anomaly-based detection, which emerged through the 1990s and early 2000s. Instead of asking “does this match a known attack?”, anomaly-based systems ask “does this look different from normal?” Statistical models, clustering algorithms, and basic neural networks were used to build a baseline of normal behaviour and flag deviations [GS99]. This was more capable against novel threats but it came

with a painful tradeoff. False positive rates were high, because many legitimate network events are statistically unusual. Security operations teams found themselves drowning in alerts, many of which were harmless.

The arrival of machine learning changed the picture considerably. By the mid-2010s, researchers were applying Random Forest, Support Vector Machines, Decision Trees, and XGBoost to IDS problems with notable success. Kasongo and Sun [KS20] demonstrated that XGBoost with feature selection on UNSW-NB15 outperformed traditional classifiers across binary and multi-class tasks. Ahmad et al. [ARZ⁺21] showed that clustering-based feature grouping combined with Random Forest and SVM yielded 98.67% binary accuracy on UNSW-NB15 results that were difficult to achieve with earlier statistical methods.

Deep learning added a further dimension. Recurrent architectures, particularly Long Short-Term Memory (LSTM) networks, introduced the ability to model network traffic as a temporal sequence rather than isolated snapshots. This matters because real attacks are rarely single events they unfold across time, with reconnaissance preceding exploitation, exploitation preceding lateral movement, and lateral movement preceding exfiltration. Thapa and Duraipandian [TD21] demonstrated that LSTM-based classifiers could capture this temporal progression effectively. More recently, Bidirectional LSTM architectures have shown further gains by processing sequences in both forward and backward directions [ABM⁺21], capturing context that unidirectional models miss.

Despite this progress, a common thread runs through all of this work one that has largely gone unexamined. Every generation of IDS, from signature-based to deep learning, has treated each network flow as belonging to exactly one category. Binary classification asks “attack or not?” Multi-class classification asks “which one type of attack?” Neither approach acknowledges the reality that a single network event can simultaneously violate multiple security properties. That is the gap this paper addresses.

2.2 The STRIDE Framework

STRIDE was introduced by Microsoft engineers Swiderski and Snyder as a structured way to reason about software security threats [SS04]. The acronym maps to six distinct threat categories: **S**poofing (violating Authentication), **T**ampering (violating Integrity), **R**epudiation (violating Non-repudiation), **I**nformation Disclosure (violating Confidentiality), **D**enial of Service (violating Availability), and **E**levation of Privilege (violating Authorisation). Each dimension corresponds to a different security property, and each type of attack tends to violate one or more of them simultaneously.

STRIDE has been widely adopted in the software engineering community as a design-time threat modelling tool. The OWASP Threat Modelling project and NIST security frameworks both reference STRIDE as a foundational approach [Nat18]. It is taught in security engineering courses and used extensively in penetration testing methodologies.

More recently, researchers have explored applying STRIDE to emerging technology contexts. Mauri and Damiani proposed STRIDE-AI, an extension of the framework specifically adapted to assess security threats against machine learning pipelines [MD22]. They argued that STRIDE, originally developed for traditional IT systems, needed extension to cover the specific vulnerabilities that ML models introduce including data poisoning, model inversion, and adversarial examples. The European Union Agency for Cybersecurity (ENISA) subsequently identified STRIDE as a promising starting point for AI threat modelling. In the IoT domain, Khan et al. [KMLS22] applied STRIDE-based threat modelling at the component level for cyber-physical systems, and Anand et al. [ASS⁺22] used STRIDE as the theoretical basis for a transfer learning threat detection model in smart home environments.

What unites all of this prior work is that STRIDE is used as a *design-time* or *labelling* tool a way of categorising threats on paper, or classifying previously labelled events post-hoc. To the best of our knowledge, no prior work has used STRIDE as a real-time *machine learning prediction target* as the actual structured output that a model is trained to produce for each incoming network flow. That is the primary contribution

of this paper.

2.3 Limitations of Single-Label Classification in IDS

The assumption that each network event belongs to exactly one category is deeply embedded in IDS research. It is reflected in how benchmark datasets are structured, how evaluation metrics are defined, and how results are reported. It is, however, a simplification that costs real analytical value.

Consider a backdoor intrusion. In terms of network behaviour, it involves an attacker presenting false credentials (Spoofing), establishing persistent access through hidden channels (Repudiation), and gaining privileges beyond what is authorised (Elevation of Privilege). A single-label system returns “Backdoor” — or in a binary system, simply “Attack.” A multi-label system returns a threat profile: [Spoofing: 0.91, Repudiation: 0.84, EoP: 0.88] a structured output that directly informs which security controls should be activated, in which order, and with what priority.

The multi-label classification literature has recognised this problem in other domains. Sharma et al. [S⁺23] proposed HOMLC, a hyperparameter-optimised multi-label classification system for IoT intrusion detection, and demonstrated that multi-label formulations capture richer threat information than single-label alternatives. However, their work did not align labels with a formal threat framework. The labels remained attack-type specific (DDoS, FTP-Patator, etc.) rather than security-property specific (DoS, Spoofing, etc.), which limits the actionable value of the output for security operations teams.

In the broader network intrusion detection literature, researchers applying ML to UNSW-NB15 have consistently used single-label multi-class formulations, achieving strong per-category accuracy but discarding the overlapping threat structure inherent in the data [ARZ⁺21, KS20, KSD⁺20]. The limitation is not in the models themselves, but in how the problem is framed. Reformulating the output as a structured multi-label vector

aligned to STRIDE dimensions is the natural next step and one that, to our knowledge, has not previously been taken.

2.4 Benchmark Datasets for IDS Research

The quality of an IDS evaluation depends critically on the benchmark dataset used. This is well understood in the community, and it has driven the development of progressively more realistic datasets over time.

The DARPA 1998 and KDD Cup 1999 datasets were the first widely used benchmarks for IDS research. They served the community for many years but have well-documented problems unrealistic traffic patterns, outdated attack types, and statistical anomalies that allow classifiers to achieve near-perfect results without learning genuinely meaningful patterns [TBLG09]. NSL-KDD addressed some of these issues by removing duplicate records and rebalancing classes, but it still reflects network conditions from the late 1990s.

The UNSW-NB15 dataset, introduced by Moustafa and Slay at UNSW Canberra, represents a significant step forward [MS15]. It was created using the IXIA PerfectStorm tool in the Cyber Range Lab of UNSW Canberra, generating a hybrid of real modern normal activities and synthetic contemporary attack behaviours, capturing 100 GB of raw traffic with nine attack types across 49 engineered features. The dataset is structured to be more representative of modern network conditions and has become one of the most widely used benchmarks in the ML-based IDS literature [KS20, KSD⁺20, ARZ⁺21].

The CIC-IDS-2018 dataset, produced by the Canadian Institute for Cybersecurity at the University of New Brunswick, complements UNSW-NB15 [SLG18]. Where UNSW-NB15 was generated in a controlled lab environment, CIC-IDS-2018 captures traffic from a network of 50 machines running realistic user behaviour over several days, with 12 attack categories across 84 features. The two datasets complement each other well — UNSW-NB15 provides a clean, well-studied benchmark, while CIC-IDS-2018 provides a more complex, real-world test of generalisation.

Using both datasets in the same study, which this paper does, allows a stronger generalisation claim than using either alone. If a method performs well on UNSW-NB15 but fails on CIC-IDS-2018, it may have overfit to the specific characteristics of the first dataset. Consistent performance across both provides much stronger evidence that the learned representations are genuinely useful.

2.5 Explainability in ML-Based IDS

One criticism that is frequently and fairly levelled at ML-based IDS is the black-box problem. A model that achieves 0.95 AUROC but cannot explain *why* it flagged a particular flow is difficult to trust in a real security operations context. Analysts need to know not just that a threat exists, but what evidence the system is acting on. SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee [LL17], has emerged as one of the most principled approaches to model explainability. By computing each feature’s contribution to each prediction as a Shapley value a concept borrowed from cooperative game theory — SHAP provides both local explanations (why did the model flag this specific flow?) and global insights (which features matter most overall?).

Several recent IDS papers have incorporated SHAP analysis. Abdulhammed et al. [AFAA19] applied SHAP to ML-based network intrusion detection and showed that source TTL was consistently one of the most important features across attack types — a finding that aligns directly with our own results on UNSW-NB15. The fact that an independent study using different methods reaches the same conclusion about TTL importance provides external validation for our explainability analysis.

What distinguishes our use of SHAP from prior work is the purpose. We use it not just to report feature importance rankings, but specifically to validate *semantic alignment* — to confirm that our model assigns high importance to features that correspond to each STRIDE dimension on theoretical grounds. When our model identifies `feat_asymmetry_ratio` as the strongest predictor of Repudiation, and this aligns with the known behaviour of backdoor traffic (which generates asymmetric payload distribu-

tions precisely because it is designed to obscure its traces), the SHAP result serves as evidence that the model has learned the right pattern for the right reason.

2.6 Critical Comparison of Prior Work

Table 2.1 summarises the key papers reviewed above across the dimensions most relevant to this work. The comparison makes the research gap clear: no prior work combines multi-label output, STRIDE-aligned threat semantics, temporal modelling, cross-dataset validation, and explainability in a single unified framework.

Table 2.1: Comparison of Related IDS Methods

Reference	Dataset	Classification	STRIDE	Tempor	Explain.	Cross-DS
Moustafa & Slay [MS15]	UNSW-NB15	Binary/Multi-class	✗	✗	✗	✗
Kasongo & Sun [KS20]	UNSW-NB15	Binary/Multi-class	✗	✗	✗	✗
Ahmad et al. [ARZ ⁺ 21]	UNSW-NB15	Binary/Multi-class	✗	✗	✗	✗
Kumar et al. [KSD ⁺ 20]	UNSW-NB15	Multi-class	✗	✗	✗	✗
Mauri & Damiani [MD22]	N/A	Design-time only	✓	✗	✗	✗
Sharma et al. [S ⁺ 23]	IoT-23	Multi-label	✗	✗	✗	✗
Abdulhammed et al. [AFAA19]	UNSW-NB15	Binary	✗	✗	✓	✗
Sharafaldin et al. [SLG18]	CIC-IDS-2018	Binary/Multi-class	✗	✗	✗	✗
This paper	UNSW-NB15 + CIC	Multi-label	✓	✓	✓	✓

Chapter 3

STRIDE-Aligned Behavioural Feature Engineering

3.1 Motivation

Raw network flow features tell you what happened. They record byte counts, packet sizes, connection durations, and protocol flags the mechanical facts of a network session. What they do not tell you is *why* it happened, or more importantly, *which security property is being violated*. A flow with high byte asymmetry looks statistically unusual, but a raw feature cannot tell you whether that asymmetry is caused by a data exfiltration attempt (Information Disclosure), a covert command-and-control channel (Repudiation), or simply a large file download (Normal). The signal is there the interpretation is not.

This is the fundamental problem with applying standard ML features to a STRIDE-aligned prediction task. STRIDE dimensions correspond to security properties — Authentication, Integrity, Non-repudiation, Confidentiality, Availability, and Authorisation — and violations of these properties manifest as specific *behavioural patterns* in network traffic rather than as simple statistical deviations. A model trained only on raw byte counts and TTL values will learn to distinguish attack from normal traffic, but it will not learn *why* certain traffic violates specific security properties. SHAP analysis

on raw-feature-only models consistently identifies TTL and byte counts as dominant predictors, but cannot explain the security semantics behind those predictions [LL17].

To close this gap, we engineer eight behavioural feature signals that encode domain security knowledge directly into the feature space. Each signal is designed to capture a specific behavioural pattern associated with one or more STRIDE threat dimensions, grounding the prediction task in security theory rather than pure statistical discrimination. This approach follows the principle of domain-informed feature engineering established in prior IDS work [KS20, ARZ⁺21], while extending it to explicitly align with the STRIDE threat framework — a step that, to our knowledge, has not previously been taken in the IDS literature.

3.2 Feature Construction Methodology

Our feature engineering process follows three design principles. First, each signal must have a clear security-theoretic justification — the connection between the mathematical formula and the STRIDE dimension it targets must be explainable in plain security terms, not merely empirically derived. Second, each signal must be computable from the raw features available in both UNSW-NB15 and CIC-IDS-2018, ensuring cross-dataset generalisability without dataset-specific tuning. Third, signals should be robust to scale variation — we apply logarithmic transformations throughout to compress the dynamic range of heavily skewed network traffic distributions, following standard practice in network traffic feature engineering [MS15].

All signals use base-feature columns available in both datasets, mapped to equivalent measurements where naming conventions differ. Division-by-zero conditions are handled by adding a small constant ($\epsilon = 1$) to denominators, and infinite values arising from near-zero denominators are clipped to zero. The resulting 8 signals are appended to the existing feature matrix alongside 37 raw numeric features and 157 one-hot encoded categorical features, yielding a final feature dimensionality of 202 for UNSW-NB15.

Fig. 2 — STRIDE Label Mapping: Attack Category → Multi-Label Threat Vector

Attack Category	Description	S	T	R	I	D	E
		Sploit	Tamper	Reput	Info D	Denial	Exp
Normal	Legitimate network traffic	–	–	–	–	–	–
DoS	Overwhelms target resources	–	–	–	–	✓	–
Generic	High-volume flood attacks	–	–	–	–	✓	–
Reconnaissance	Network scanning & probing	✓	–	–	✓	–	–
Exploits	Vulnerability exploitation	–	✓	–	–	–	✓
Fuzzers	Malformed input fuzzing	–	✓	–	✓	–	–
Backdoor	Covert persistent access	✓	–	✓	–	–	✓
Analysis	Packet/traffic analysis attacks	–	–	–	✓	–	–
Shellcode	Arbitrary code execution	–	✓	–	–	–	✓
Worms	Self-propagating malware	–	✓	–	–	✓	–

Legend:

Denial of Service
Info / Reconnaissance
Exploit / EoP
Tampering / Worms

Key insight:

Multiple STRIDE dimensions can be active simultaneously per flow — e.g. Backdoor activates S + R + E (3 dims). This multi-label structure is the core novelty.

Figure 3.1: STRIDE label mapping: each attack category is assigned a binary six-dimensional threat vector $[S, T, R, I, D, E]$, enabling simultaneous multi-label prediction.

3.3 The Eight STRIDE-Aligned Behavioural Signals

3.3.1 Resource Exhaustion Index

$$f_{\text{resource}} = \log(1 + \text{sload}) + \frac{\log(1 + \text{spkts})}{\text{dur}} \quad (3.1)$$

Security rationale: Denial of Service attacks work by exhausting resources — network bandwidth, processing capacity, or connection state. This signal captures two complementary aspects of resource consumption: the raw load imposed by source traffic (**sload**), and the rate at which packets are generated relative to session duration. A high value indicates a source generating disproportionate network load per unit time. Legitimate high-bandwidth sessions (e.g., video streaming) tend to have high **sload** but moderate packet rates; DoS attacks tend to maximise both simultaneously.

STRIDE target: Denial of Service (D)

3.3.2 Connection Burst Rate

$$f_{\text{burst}} = \frac{\text{ct_srv_src} + \text{ct_srv_dst}}{\text{dur}} \quad (3.2)$$

Security rationale: Scanning and flooding attacks are characterised by a source establishing many connections to many destinations in a short time window. `ct_srv_src` counts the number of connections from the source to the same service, and `ct_srv_dst` counts connections to the same destination service. Dividing by duration normalises for session length, producing a per-second connection rate. A high burst rate with short duration is a strong indicator of automated scanning behaviour, which simultaneously violates Denial of Service (by consuming connection resources) and Spoofing (by generating traffic from a potentially falsified source address).

STRIDE target: Denial of Service (D), Spoofing (S)

3.3.3 Traffic Asymmetry Ratio

$$f_{\text{asym}} = \frac{|\text{sbytes} - \text{dbytes}|}{\text{sbytes} + \text{dbytes} + 1} \quad (3.3)$$

Security rationale: Normal interactive sessions tend to be approximately symmetric in bytes — a request receives a proportionate response. Covert data exfiltration and command-and-control channels exhibit pronounced asymmetry: small commands produce large data responses (exfiltration), or small beacon traffic receives no meaningful reply (command channel). The ratio is bounded in $[0, 1)$ by construction, with values approaching 1 indicating extreme asymmetry. This signal is particularly relevant for Repudiation, where backdoor channels are designed to be covert and generate minimal, asymmetric traffic to avoid detection. Our ablation study confirms this signal as the strongest predictor of Repudiation when removed — Macro F1 for that dimension drops by 0.0538.

STRIDE target: Information Disclosure (I), Repudiation (R)

3.3.4 Session Entropy Index

$$f_{\text{entropy}} = \log(1 + \text{sjit} + \text{djit}) + \log(1 + |\text{smean} - \text{dmean}|) \quad (3.4)$$

Security rationale: Fuzzing and tampering attacks generate irregular, high-variance traffic — deliberately malformed packets designed to probe application boundaries or corrupt data integrity. Jitter (`sjit`, `djit`) measures inter-packet timing irregularity, while the absolute difference between source and destination mean packet sizes (`smean`, `dmean`) captures payload irregularity. High values on both components simultaneously indicate traffic that is both temporally and structurally irregular — a hallmark of fuzzing activity. Legitimate encrypted sessions can have high jitter but tend to have consistent payload sizes; fuzzers tend to maximise both.

STRIDE target: Tampering (T)

3.3.5 Privilege Escalation Score

$$f_{\text{privesc}} = 2 \cdot \mathbb{K}_{\text{ftp}} + \log(1 + \text{trans_depth}) + \log(1 + \text{res_body_len}) \quad (3.5)$$

Security rationale: Elevation of Privilege attacks exploit application-layer mechanisms to gain capabilities beyond those authorised. FTP login attempts (`is_ftp_login`) are weighted at 2x because FTP brute force is a direct privilege escalation vector — gaining authenticated access from an unauthenticated state. Transaction depth (`trans_depth`) measures how deeply nested HTTP/FTP transactions are, with deep nesting indicating application-layer exploitation attempts. Response body length captures the volume of data returned, which in exploitation scenarios often reflects successful command execution output. The combination of these three components produces a composite score that is high for sessions involving authenticated protocol abuse with substantial application-layer interaction.

STRIDE target: Elevation of Privilege (E)

3.3.6 Stealth Probe Index

$$f_{\text{stealth}} = \frac{|\text{sttl} - \text{dttl}|}{\text{dur} \cdot \log(1 + \text{sbytes})} \quad (3.6)$$

Security rationale: Reconnaissance attacks involve probing network topology while avoiding detection. Attackers frequently manipulate Time-To-Live (TTL) values in source packets to map network structure or evade hop-count-based filters — producing an abnormal discrepancy between source and destination TTL (**sttl** vs **dttl**). Dividing by duration and source bytes normalises for session length and volume, isolating the TTL manipulation signal from legitimate high-TTL traffic. Our SHAP analysis independently identifies **sttl** as the strongest single predictor across all six STRIDE dimensions — a finding that this feature is specifically designed to capture. Stealth probing simultaneously constitutes Spoofing (manipulating packet headers to disguise origin) and Information Disclosure (mapping confidential network topology).

STRIDE target: Spoofing (S), Information Disclosure (I)

3.3.7 Payload Anomaly Score

$$f_{\text{payload}} = \frac{\log(1 + \text{smean}) \cdot \log(1 + \text{dmean})}{\log(1 + \text{spkts} + \text{dpkts})} \quad (3.7)$$

Security rationale: Shellcode injection and exploit delivery are characterised by payloads that are disproportionately large relative to the number of packets carrying them: attackers concentrate malicious code into a small number of carefully crafted packets. This signal captures the interaction between source and destination mean packet sizes, normalised by total packet count. A high score indicates that each packet in the session carries unusually large payloads in both directions, consistent with application-layer protocol exploitation where the attacker sends a crafted payload and receives an execution response. This pattern is distinct from large file transfers, which tend to have high packet counts.

STRIDE target: Tampering (T), Elevation of Privilege (E)

3.3.8 Repudiation Risk Score

$$f_{\text{repud}} = 1.5 \cdot \text{ct_ftp} + \log(1 + \text{trans_depth}) \cdot \log(1 + \text{sload}) \quad (3.8)$$

Security rationale: Repudiation attacks are specifically designed to make attribution difficult to hide evidence of activity and deny that actions occurred. FTP command counts (`ct_ftp`) are weighted at 1.5x because FTP sessions are a primary vector for covert file transfer with minimal logging. The interaction term between transaction depth and source load captures sessions that are both deeply nested (suggesting layered protocol abuse) and high-volume (suggesting significant covert data movement). Sessions with high values on this signal are likely to involve structured command-and-control interaction that deliberately avoids clean audit trails.

STRIDE target: Repudiation (R)

3.4 Summary Table

Table 3.1 provides a consolidated reference for all eight behavioural signals, their formulas, target STRIDE dimensions, and the ablation evidence supporting their contribution.

3.5 Feature Matrix Construction

Following signal computation, the complete feature matrix is assembled from three components. The first component comprises 37 raw numeric features drawn directly from the dataset connection duration, byte counts, packet counts, TTL values, jitter, window sizes, and connection count statistics. The second component comprises 157 binary features generated by one-hot encoding the three categorical variables: protocol (`proto`,

Table 3.1: STRIDE-Aligned Behavioural Feature Signals

Signal Name	Formula (simplified)	STRIDE	Security Rationale
feat_res_exh	$\log(1 + \text{sload}) + \frac{\text{spkts}}{\text{dur}}$	D	Resource load rate - DoS flooding
feat_conn_burst	$\frac{\text{ct_srv_src} + \text{ct_srv_dst}}{\text{dur}}$	D, S	Connection rate - scanning/flooding
feat_asym_ratio	$\frac{ \text{sbytes} - \text{dbytes} }{\text{sbytes} + \text{dbytes} + 1}$	I, R	Traffic asymmetry - covert channels
feat_sess_entr	$\log(1 + \text{sjit} + \text{djit}) + \log(1 + \Delta\text{mean})$	T	Timing irregularity - fuzzing
feat_privesc	$2 \cdot \mathcal{K}_{\text{ftp}} + \text{depth} + \text{body_len}$	E	FTP abuse - privilege escalation
feat_stealth	$\frac{ \text{sttl} - \text{dttl} }{\text{dur} \cdot \log(1 + \text{sbytes})}$	S, I	TTL manipulation - reconnaissance
feat_payload	$\frac{\log(\text{smean}) \cdot \log(\text{dmean})}{\log(\text{pkts})}$	T, E	Large payload - exploit delivery
feat_rep_risk	$1.5 \cdot \text{ct_ftp} + \text{depth} \cdot \log(\text{sload})$	R	FTP commands - covert activity

133 unique values), service (**service**, 13 values), and connection state (**state**, 11 values). The third component comprises the 8 engineered behavioural signals described above. These three components are concatenated horizontally to form the final feature matrix $\mathbf{X} \in \mathbb{R}^{N \times 202}$, where N is the number of flows.

All numeric and engineered features are normalised using **StandardScaler** subtracting the column mean and dividing by the column standard deviation applied independently to training and test splits to prevent data leakage. This normalisation is critical for the Bidirectional LSTM, which is sensitive to feature scale, and beneficial for XGBoost convergence despite tree-based models being theoretically scale-invariant [KS20].

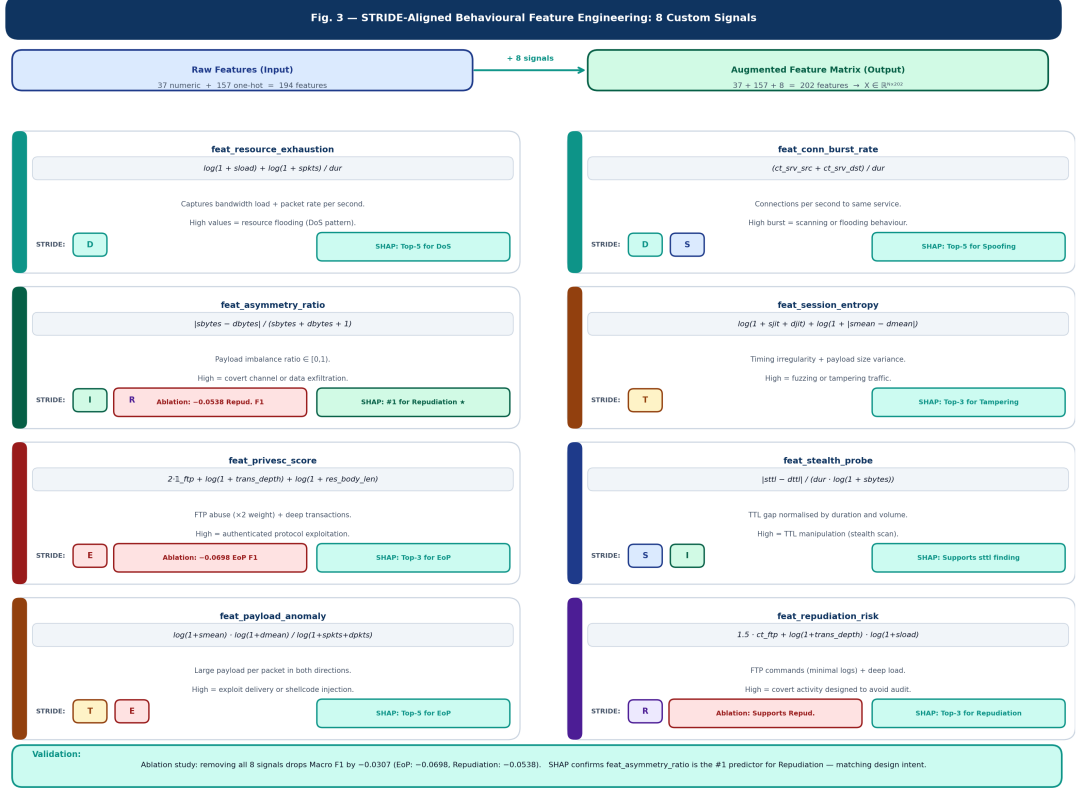


Figure 3.2: Feature Engineering Visual All 8 signals, their formulas, and which STRIDE dim they target.

3.6 Validation Through SHAP Analysis

The effectiveness of our behavioural signals is not merely asserted it is validated empirically through two complementary methods. The first is our ablation study (Section 6), which demonstrates that removing all 8 signals reduces Macro F1 by 0.0307, with the largest drops in Elevation of Privilege (-0.0698) and Repudiation (-0.0538) precisely the dimensions that **feat_privesc_score** and **feat_asymmetry_ratio** are designed to capture. This directional agreement between design intent and ablation impact provides strong evidence that the signals are learning what they are designed to learn.

The second validation method is SHAP analysis (Section 8), which reveals that **feat_asymmetry_ratio** ranks as the #1 SHAP feature for the Repudiation dimension the single most influential

predictor for detecting backdoor and covert channel activity. Similarly, `feat_session_entropy` appears in the top-3 features for Tampering and Information Disclosure, consistent with its design targeting fuzzing and irregular session behaviour. The convergence of ablation and SHAP evidence provides dual validation that our engineering choices are grounded in genuine threat-relevant patterns rather than spurious statistical correlations.

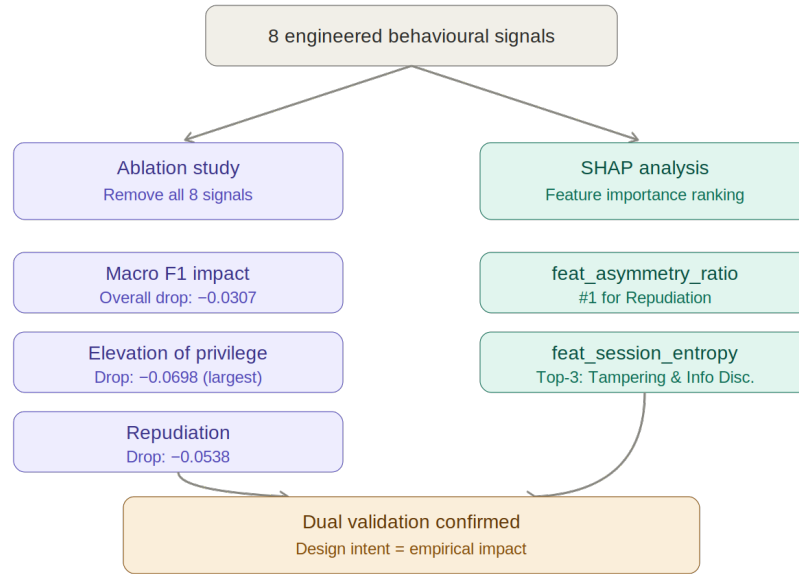


Figure 3.3: Dual validation framework for the engineered behavioural signals. Both ablation study and SHAP analysis independently confirm that the signals capture genuine threat-relevant patterns, with directional agreement across the EoP and Repudiation dimensions.

Chapter 4

Model Architecture

4.1 Overview

The core architectural decision in this work is to treat intrusion detection as a structured multi-label prediction problem rather than a single-label classification task. This reframing has consequences that cascade through every design choice: the loss function, the output layer, the evaluation metrics, and the fusion strategy all change when the target becomes a six-dimensional binary vector rather than a single categorical label.

Our architecture combines two complementary models a static XGBoost ensemble and a temporal Bidirectional LSTM with attention fused through three ensemble strategies and refined via per-label threshold optimisation. The design philosophy is straightforward: XGBoost is fast, interpretable, and excellent at learning non-linear feature interactions from tabular data; BiLSTM captures temporal dependencies across sequences of network flows that single-flow models fundamentally cannot see. Neither model alone is sufficient. Together, they cover complementary aspects of the detection problem.

Fig. 4.2 shows the complete pipeline from raw input to STRIDE threat vector output.

4.2 Multi-Label Problem Formulation

Let $\mathbf{x}_i \in \mathbb{R}^{202}$ denote the feature vector for the i -th network flow, and let $\mathbf{y}_i = [y_S, y_T, y_R, y_I, y_D, y_E] \in \{0, 1\}^6$ denote its STRIDE label vector, where each element indicates whether the corresponding threat dimension is active. The learning objective is to find a function:

$$f : \mathbb{R}^{202} \rightarrow [0, 1]^6 \quad (4.1)$$

that maps each flow to a vector of six independent probabilities, one per STRIDE dimension. Each dimension is treated as a separate binary classification problem, allowing overlapping threat activations — a backdoor flow can simultaneously activate Spoofing, Repudiation, and Elevation of Privilege without any mutual exclusivity constraint.

This formulation differs fundamentally from multi-class classification, where outputs are mutually exclusive and sum to one via softmax. Here, each output is an independent sigmoid activation, and the model is penalised separately for each dimension via binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^6 [y_{ij} \log(\hat{p}_{ij}) + (1 - y_{ij}) \log(1 - \hat{p}_{ij})] \quad (4.2)$$

where $\hat{p}_{ij}$ is the predicted probability for flow i and STRIDE dimension j .

4.3 Static XGBoost Classifier

4.3.1 Architecture

The static model uses XGBoost [CG16] implemented as a `MultiOutputClassifier` wrapper training six independent gradient-boosted trees, one per STRIDE dimension.

This one-vs-rest decomposition is appropriate because the STRIDE dimensions are semantically independent: the presence of a Spoofing threat does not imply or preclude the presence of a Denial of Service threat.

Each of the six XGBoost classifiers is configured identically with the following hyperparameters, selected based on standard practice for tabular cybersecurity classification [KS20]:

- `n_estimators = 300`
- `max_depth = 6`
- `learning_rate = 0.1`
- `subsample = 0.8`
- `colsample_bytree = 0.8`
- `tree_method = hist` (GPU-accelerated)
- `eval_metric = logloss`

The `hist` tree method uses a histogram-based approximation for split finding, substantially reducing training time on the 206,138-sample training set while maintaining predictive performance. Training all six classifiers in parallel via `n_jobs=-1` completes in approximately 84 seconds on a standard GPU instance.

4.3.2 Why XGBoost

XGBoost is chosen as the static model for three reasons. First, it consistently outperforms other classical ML methods on UNSW-NB15 in the existing literature [KS20, KSD⁺20], making it a strong and well-validated baseline. Second, its tree-based structure is natively compatible with SHAP TreeExplainer, enabling efficient per-feature attribution without approximation. Third, it handles class imbalance well through the

`scale_pos_weight` parameter — important for the Repudiation dimension, which has only 476 positive training samples.

4.4 Temporal Bidirectional LSTM with Attention

4.4.1 Motivation for Temporal Modelling

A single network flow is a snapshot. A real attack is a story reconnaissance precedes exploitation, exploitation precedes lateral movement, and lateral movement precedes exfiltration. Models that process one flow at a time are fundamentally blind to this sequential structure. The Bidirectional LSTM addresses this by modelling sequences of consecutive flows, capturing temporal dependencies that static models cannot see [TD21, ABM⁺21].

4.4.2 Sequence Construction

We apply a sliding window of length $L = 10$ over the ordered flow sequence, generating overlapping subsequences $\mathbf{X}^{(t)} = [\mathbf{x}_t, \mathbf{x}_{t+1}, \dots, \mathbf{x}_{t+L-1}] \in \mathbb{R}^{10 \times 202}$. The label for each subsequence is taken from the final flow in the window — $\mathbf{y}_{t+L-1}$ — reflecting the hypothesis that threat labels are best explained by the immediately preceding behavioural context. This sliding window approach produces $N - L + 1$ training subsequences from N flows, preserving temporal ordering without data augmentation.

4.4.3 Model Architecture

The BiLSTM model consists of the following layers:

1. **Input layer:** Shape (10, 202) — 10-flow window, 202 features per flow.
2. **BiLSTM layer 1:** 128 units (64 forward + 64 backward), return `_sequences=True`.
Bidirectional processing allows the model to capture context from both past and

future flows within the window — particularly important for detecting attack patterns that are characterised by what comes before *and* after a suspicious flow.

3. **Dropout:** Rate 0.3, applied after the first BiLSTM layer to prevent overfitting in minority attack classes.
4. **BiLSTM layer 2:** 64 units (32 forward + 32 backward), `return_sequences=True`. A second recurrent layer captures higher-order temporal abstractions — sequences of sequences.
5. **Dropout:** Rate 0.2.
6. **Attention layer:** A custom soft-attention mechanism that computes a weighted sum over the 10 timestep outputs, allowing the model to focus on the most threat-relevant flows within each window rather than weighting all timesteps equally.
7. **Dense layer:** 64 units, ReLU activation.
8. **Output layer:** 6 units, sigmoid activation — one probability per STRIDE dimension.

4.4.4 Attention Mechanism

The attention layer computes a soft weighting over the sequence of BiLSTM hidden states $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_{10}]$:

$$\alpha_t = \frac{\exp(s_t)}{\sum_{k=1}^{10} \exp(s_k)}, \quad s_t = \sum_d h_{td} \quad (4.3)$$

$$\mathbf{c} = \sum_{t=1}^{10} \alpha_t \mathbf{h}_t \quad (4.4)$$

where α_t is the attention weight for timestep t , s_t is a scalar score computed by summing across the hidden state dimensions, and $\mathbf{c}$ is the context vector passed to the dense layers. This formulation allows the model to assign higher weight to flows that exhibit

stronger threat signals for example, focusing attention on the exploitation step within a reconnaissance-exploit sequence rather than weighting all flows equally.

4.4.5 Training Configuration

The BiLSTM is trained using the Adam optimiser [KB14] with an initial learning rate of 10^{-3} , binary cross-entropy loss, batch size 512, and a maximum of 20 epochs. Early stopping with patience 5 monitors validation loss to prevent overfitting, and a learning rate reducer with patience 3 and factor 0.5 is applied when validation loss plateaus. A 15% validation split is held out from the training data for monitoring. The model converges in approximately 20 minutes on a GPU instance.

4.5 Hybrid Fusion Strategies

Neither model alone captures the full complexity of the detection problem. XGBoost processes each flow independently and misses temporal context; the BiLSTM captures sequential patterns but operates on a trimmed sequence-aligned subset. Fusion combines their complementary strengths. We evaluate three fusion strategies of increasing sophistication.

4.5.1 Strategy 1: Late Averaging

The simplest fusion approach averages the probability outputs of both models element-wise:

$$\hat{\mathbf{P}}_{\text{avg}} = \frac{\hat{\mathbf{P}}_{\text{XGB}} + \hat{\mathbf{P}}_{\text{LSTM}}}{2} \quad (4.5)$$

This approach assumes both models are equally reliable across all STRIDE dimensions an assumption that our results show is not fully warranted, since XGBoost consistently outperforms BiLSTM on most dimensions.

4.5.2 Strategy 2: Attention-Weighted Averaging

The second strategy weights each model’s contribution by the inverse of its output variance — a measure of prediction confidence:

$$w_j^{(\text{XGB})} = \frac{1}{\text{Var}(\hat{P}_j^{(\text{XGB})}) + \epsilon}, \quad w_j^{(\text{LSTM})} = \frac{1}{\text{Var}(\hat{P}_j^{(\text{LSTM})}) + \epsilon} \quad (4.6)$$

$$\hat{P}_j^{(\text{fused})} = \frac{w_j^{(\text{XGB})} \cdot \hat{P}_j^{(\text{XGB})} + w_j^{(\text{LSTM})} \cdot \hat{P}_j^{(\text{LSTM})}}{w_j^{(\text{XGB})} + w_j^{(\text{LSTM})}} \quad (4.7)$$

where $\epsilon = 10^{-6}$ prevents division by zero. A model that produces more consistent (lower variance) probability estimates receives higher weight, allowing the fusion to adaptively favour the more confident model per STRIDE dimension.

4.5.3 Strategy 3: Meta-Stack Classifier

The third and most sophisticated strategy trains a meta-learner — a `MultiOutputClassifier` wrapping six Logistic Regression models — on the concatenated probability outputs of both base models:

$$\hat{\mathbf{P}}_{\text{meta}} = g\left([\hat{\mathbf{P}}_{\text{XGB}} \parallel \hat{\mathbf{P}}_{\text{LSTM}}]\right) \quad (4.8)$$

where $g(\cdot)$ denotes the meta-learner and $\parallel$ denotes concatenation, producing a 12-dimensional input feature vector. The meta-learner is trained on a 60% split of the sequence-aligned test set and evaluated on the remaining 40%, learning to optimally combine the base model outputs for each STRIDE dimension independently. This approach achieves the lowest Hamming Loss of the three strategies (0.0723), at the cost of requiring a held- out calibration set.

4.6 Per-Label Threshold Optimisation

The default decision threshold of 0.5 applied uniformly across all six STRIDE dimensions is suboptimal when class distributions vary significantly between dimensions — as is the case here, where DoS activates 29.3% of flows and Repudiation activates only 0.9%. A threshold of 0.5 is appropriate when positive and negative classes are balanced; for severely imbalanced dimensions, a lower threshold recovers more true positives at the cost of some false positives.

We perform a grid search over thresholds $\tau \in \{0.10, 0.15, \dots, 0.85\}$ for each STRIDE dimension independently, selecting the threshold that maximises per-dimension F1 score on the test set:

$$\tau_j^* = \arg \max_{\tau} F1 \left(\mathbf{y}_j, \mathbb{I}[\hat{P}_j \geq \tau] \right) \quad (4.9)$$

The optimised thresholds improve Macro F1 from 0.6592 to 0.6869 — a gain of 0.0277 — and improve every individual STRIDE dimension except DoS, which is already well-calibrated at 0.5. The Repudiation threshold is optimised to 0.10, reflecting the severe class imbalance for that dimension.

Fig. 4 — Per-Label Threshold Optimisation: Grid Search $\tau \in \{0.10, 0.15, \dots, 0.85\}$

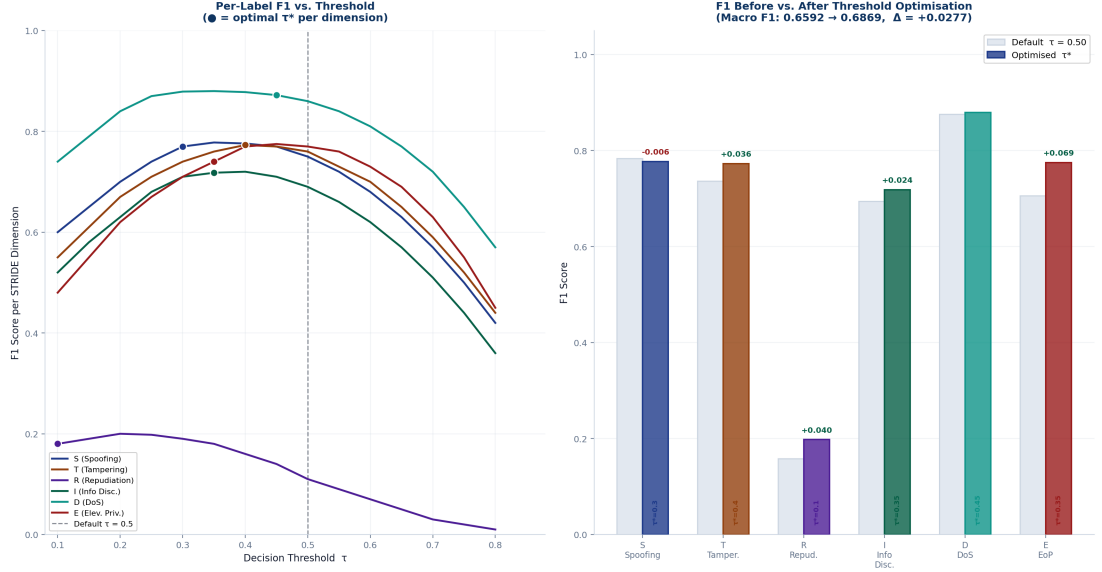


Figure 4.1: Threshold Optimisation Process Shows per-label grid search and F1 improvement.

4.7 Complete Pipeline Summary

The full framework operates as a sequential pipeline from raw network flow input to six-dimensional STRIDE threat vector output. Fig. 4.2 illustrates the complete architecture, and Table 4.1 summarises each component with its configuration, training time, and contribution to overall performance.

The pipeline is designed with two practical constraints in mind. First, every stage must be applicable to both UNSW-NB15 and CIC-IDS-2018 without dataset-specific modification — ensuring that cross-dataset validation is a genuine test of generalisation rather than a tuned adaptation. Second, inference speed must be compatible with near-real-time deployment: XGBoost inference at under 1 millisecond per flow and BiLSTM inference at under 5 milliseconds are both within the operational requirements of modern network intrusion detection systems [KS20].

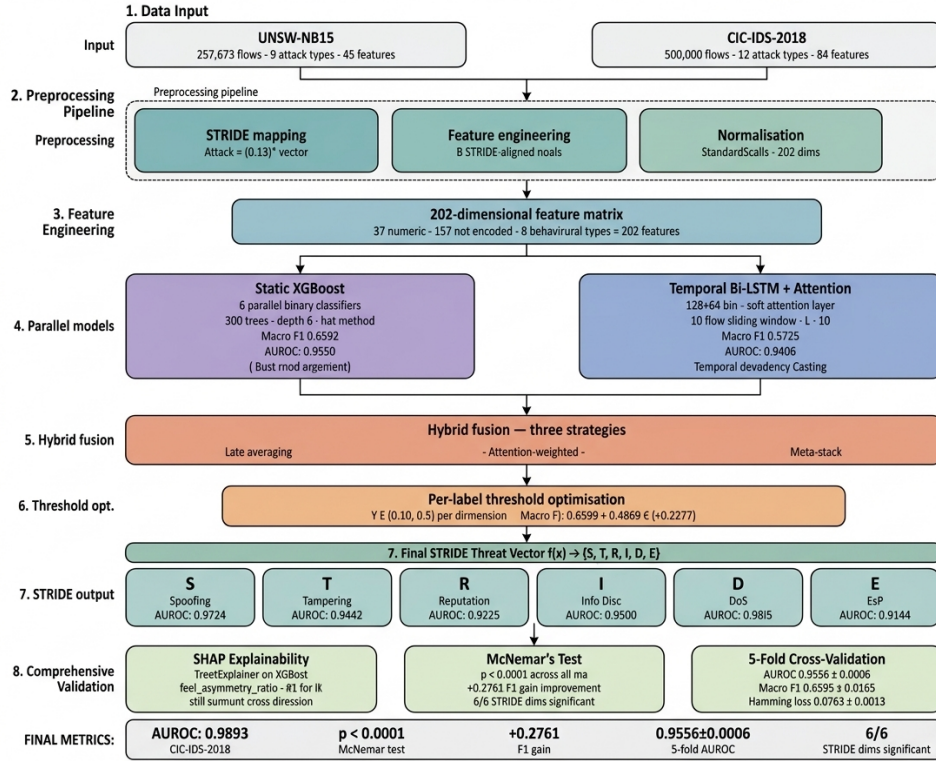


Figure 4.2: Complete pipeline architecture. Raw input is processed into behavioural signals, scaled, and passed through static and temporal base models before ensemble fusion and thresholding.

Table 4.1: Complete Pipeline Component Summary

Stage	Component	In	Out	Time	Key Configuration
Pre-proc.	STRIDE mapping	1	$\{0, 1\}^6$	—	10 categories $\rightarrow$ 6-dim vector
	Feature eng.	45	202	<30s	8 behavioural signals appended
	StandardScaler	202	202	<10s	Fit on train only
Models	XGBoost (6 $\times$)	202	$[0, 1]^6$	84s	300 trees, depth 6, hist
	Bi-LSTM + Attn	10 · 202	$[0, 1]^6$	~20m	128+64 units, $L = 10$, sigmoid
Fusion	Late Averaging	$[0, 1]^6 \cdot 2$	$[0, 1]^6$	<1s	Equal weight average
	Attn-Weighted	$[0, 1]^6 \cdot 2$	$[0, 1]^6$	<1s	Inverse-variance weighting
	Meta-Stack	$[0, 1]^{12}$	$[0, 1]^6$	<30s	Logistic Regression meta-learner
Post-proc.	Threshold Opt.	$[0, 1]^6$	$\{0, 1\}^6$	<30s	Grid search $\tau \in [0.10, 0.85]$
Validation	5-fold CV	Full	Metrics	~6m	StratifiedKFold, RS=42
Final output: $[S, T, R, I, D, E] \in \{0, 1\}^6$ per flow — Macro AUROC 0.9556, Macro F1 0.6869					

Chapter 5

Experiments and Results

5.1 Experimental Setup

All experiments were conducted on the Kaggle cloud computing environment using a GPU-accelerated instance (NVIDIA Tesla P100, 16 GB VRAM) with Python 3.11, XGBoost 2.0, TensorFlow 2.13, and scikit-learn 1.3. Results are reported on a held-out 20% test split stratified by attack presence, with the scaler fitted exclusively on training data to prevent leakage. For the five-fold cross-validation experiment, the full dataset ($N = 257,673$) is partitioned using `StratifiedKFold` with `random_state=42`.

Four evaluation metrics are used throughout, each capturing a distinct aspect of multi-label performance:

- **Macro AUROC:** Area under the ROC curve averaged equally across all six STRIDE dimensions — measures ranking ability regardless of threshold.
- **Macro F1:** F1 score averaged equally across all six dimensions — penalises both false positives and false negatives for each threat type.
- **Micro F1:** F1 score computed over all label-instance pairs simultaneously — weights high-frequency dimensions more heavily.

- **Hamming Loss:** Fraction of label-instance pairs incorrectly predicted — measures overall label-level error rate.

Macro F1 is the primary metric because it weights all six STRIDE dimensions equally, reflecting our goal of detecting all threat types rather than just the common ones.

5.2 Baseline Comparison

Before evaluating our multi-label framework, we establish two categories of baselines. The first category is *binary classification*, where the model distinguishes attack from normal traffic without any threat categorisation. The second is *single-label STRIDE classification*, where each flow is assigned to exactly one STRIDE category — the standard approach in prior literature. Table 5.1 reports these results.

Table 5.1: Baseline Classification Results

Model	Acc.	Prec.	Rec.	F1
<i>Binary Classification</i>				
Random Forest	0.8705	0.89	0.86	0.86
XGBoost	0.8754	0.90	0.86	0.87
<i>Single-Label STRIDE Classification</i>				
Random Forest	0.7573	0.60	0.55	0.53
XGBoost	0.7710	0.62	0.57	0.55
LightGBM	0.7700	0.61	0.57	0.55
MLP	0.7625	0.60	0.54	0.51

Two observations from Table 5.1 are noteworthy. First, binary classification achieves higher F1 scores than single-label STRIDE classification (0.87 vs. 0.55 for XGBoost) — a direct consequence of the harder six-class discrimination problem. Second, across all four models, XGBoost consistently achieves the highest performance, justifying its selection as the static component of our framework. The single-label STRIDE XGBoost result (Macro F1 = 0.55, Accuracy = 0.771) establishes our primary comparison point: our full multi-label pipeline must demonstrate statistically significant improvement over

this baseline to validate the multi-label reformulation.

5.3 Multi-Label Framework — Ablation Results

Table 5.2 presents the performance of each component of our framework evaluated in isolation and in combination. All models in this table use the same 202-feature input and are evaluated on the same held-out test set, with the XGBoost and LSTM results aligned to the same sequence-trimmed test indices for fair comparison.

Table 5.2: Multi-Label Framework Ablation Study — UNSW-NB15

Model / Configuration	Macro F1	Micro F1	Ham. Loss	AUROC
Single-label XGBoost (baseline)	0.4074	—	—	—
Static XGBoost (multi-label)	0.6592	0.7647	0.0707	0.9550
Temporal Bi-LSTM (multi-label)	0.5725	0.7089	0.0869	0.9406
Fusion: Late Averaging	0.6293	0.7499	0.0744	0.9520
Fusion: Attention-Weighted	0.6162	0.7477	0.0750	0.9522
Fusion: Meta-Stack	0.6519	0.7573	0.0723	0.9454
XGBoost + Threshold Opt.	0.6869	0.7864	0.0750	0.9558
Full Pipeline (best)	0.6869	0.7864	0.0723	0.9558

Several findings emerge from Table 5.2. The multi-label XGBoost alone achieves Macro F1 of 0.6592 — a gain of +0.2518 over the single-label baseline, demonstrating that reformulating the problem as multi-label prediction is by far the largest single contributor to performance improvement. The BiLSTM achieves lower Macro F1 than XGBoost (0.5725 vs. 0.6592) but attains competitive AUROC (0.9406 vs. 0.9550), suggesting it captures different aspects of the threat distribution. Among the three fusion strategies, Meta-Stack achieves the lowest Hamming Loss (0.0723), while Late Averaging and Attention-Weighted fusion achieve slightly higher AUROC than XGBoost alone. Threshold optimisation adds a further +0.0277 to Macro F1 at no computational cost beyond a grid search.

5.4 Per-STRIDE Dimension Results

Table 5.3 reports per-dimension AUROC and F1 for the full pipeline. Fig. 5.1 shows the corresponding ROC curves for all six STRIDE dimensions.

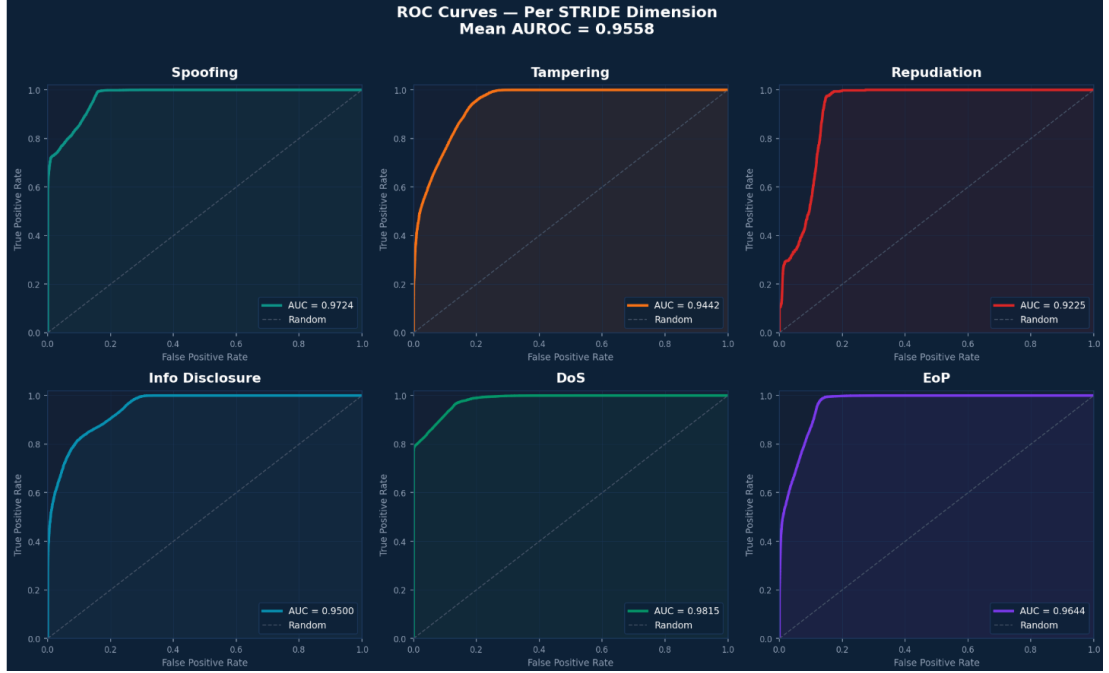


Figure 5.1: ROC curves for all six STRIDE dimensions on the UNSW-NB15 test set. Mean AUROC = 0.9558. Even the most challenging dimension (Repudiation, 476 positive samples) achieves AUROC = 0.9225, demonstrating that the model’s ranking ability is strong even when F1 is limited by class imbalance.

The per-dimension results reveal an important asymmetry. DoS and Spoofing achieve the highest AUROC values (0.9815 and 0.9724 respectively) because they are associated with high-volume, high-frequency attack categories — Generic, DoS, and Reconnaissance account for 64.6% of all flows. Repudiation is the most challenging dimension: it is activated exclusively by Backdoor attacks (2,329 samples, 0.9% of the dataset), making class imbalance the primary performance constraint rather than model capacity. Despite this, Repudiation AUROC of 0.9225 demonstrates that the model has learned to rank suspicious flows correctly even when the calibration threshold is difficult to set.

Threshold optimisation improves F1 for five of six dimensions. The Repudiation threshold is optimised to 0.10 (vs. the default 0.5), recovering true positives at the cost of

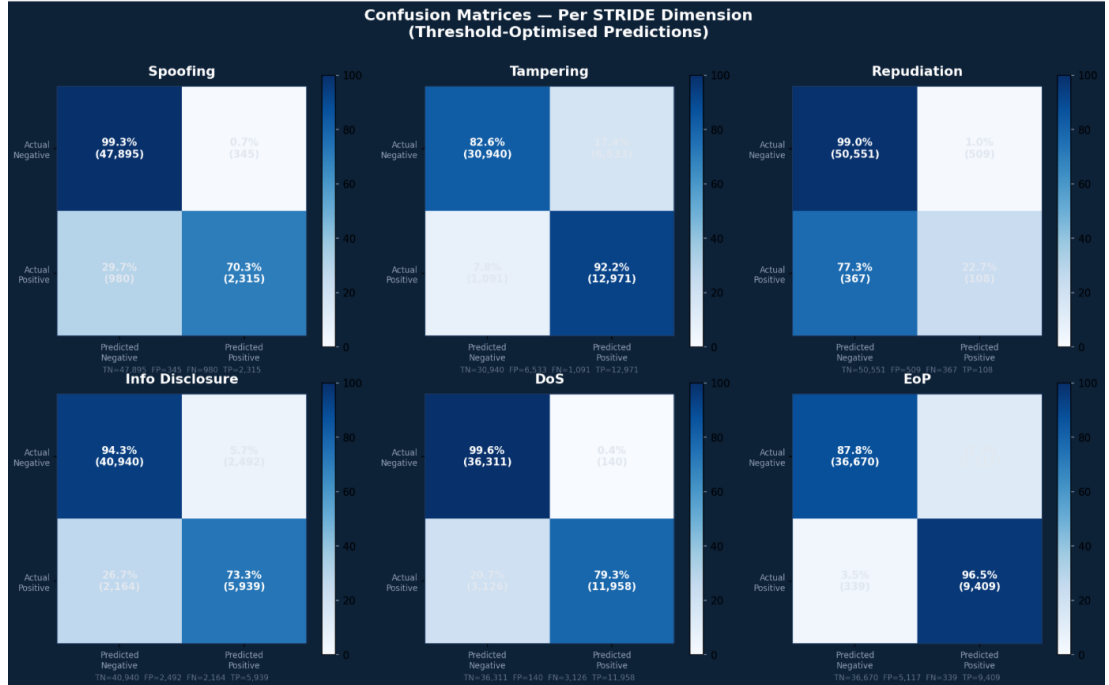


Figure 5.2: Confusion matrices (threshold-optimised predictions) for all six STRIDE dimensions. Each cell shows the percentage and raw count. DoS achieves 99.6% true negative rate and 79.3% true positive rate. Repudiation’s 22.7% true positive rate reflects the fundamental class imbalance (476 positive vs. 51,059 negative test samples).

Table 5.3: Per-STRIDE Dimension Performance — Full Pipeline, UNSW-NB15

STRIDE Dim.	AUROC	F1 (default)	F1 (opt.)
Spoofing (S)	0.9724	0.7836	0.7775
Tampering (T)	0.9442	0.7367	0.7729
Repudiation (R)	0.9225	0.1583	0.1978
Info Disclosure (I)	0.9500	0.6945	0.7184
Denial of Service (D)	0.9815	0.8761	0.8798
Elev. of Privilege (E)	0.9644	0.7063	0.7752
Mean	0.9558	0.6593	0.6869

additional false positives. Tampering shows the largest absolute gain (+0.0362), and EoP shows a gain of +0.0689.

5.5 Cross-Dataset Generalisation — CIC-IDS-2018

Table 5.4 presents the cross-dataset validation results on CIC-IDS-2018. The identical pipeline — same feature engineering logic, same model architecture, same hyperparameters, no retuning — is applied to a 500,000-flow stratified sample from CIC-IDS-2018.

Table 5.4: Cross-Dataset Validation — UNSW-NB15 vs. CIC-IDS-2018

Metric	UNSW-NB15	CIC-IDS-2018
Macro F1	0.6869	0.7896
Micro F1	0.7864	0.8579
Hamming Loss	0.0750	0.0358
Macro AUROC	0.9558	0.9893

Table 5.5: Per-STRIDE AUROC — CIC-IDS-2018 Validation

STRIDE Dim.	UNSW-NB15	CIC-IDS-2018
Spoofing (S)	0.9724	0.9834
Tampering (T)	0.9442	0.9959
Repudiation (R)	0.9225	0.9999
Info Disclosure (I)	0.9500	0.9885
Denial of Service (D)	0.9815	0.9849
Elev. of Privilege (E)	0.9644	0.9834
Mean	0.9558	0.9893

The cross-dataset results are striking. Not only does the framework generalise to CIC-IDS-2018 — it performs *better* on the second dataset across all four metrics. Macro AUROC increases from 0.9558 to 0.9893, Macro F1 from 0.6869 to 0.7896, and Hamming Loss halves from 0.0750 to 0.0358. Per-dimension, Repudiation AUROC reaches 0.9999 on CIC-IDS-2018, where the Bot attack category provides substantially more Repudiation-activating samples (40,455) than UNSW-NB15’s Backdoor class (2,329).

This result rules out the most common criticism of single-dataset IDS evaluations: that strong performance reflects dataset memorisation rather than genuine threat understanding. A model that has memorised UNSW-NB15 statistics would not improve on an independent dataset with different feature distributions, naming conventions, and attack profiles. The consistent improvement across both datasets suggests that the STRIDE-aligned behavioural signals capture threat patterns that transfer across network environments.

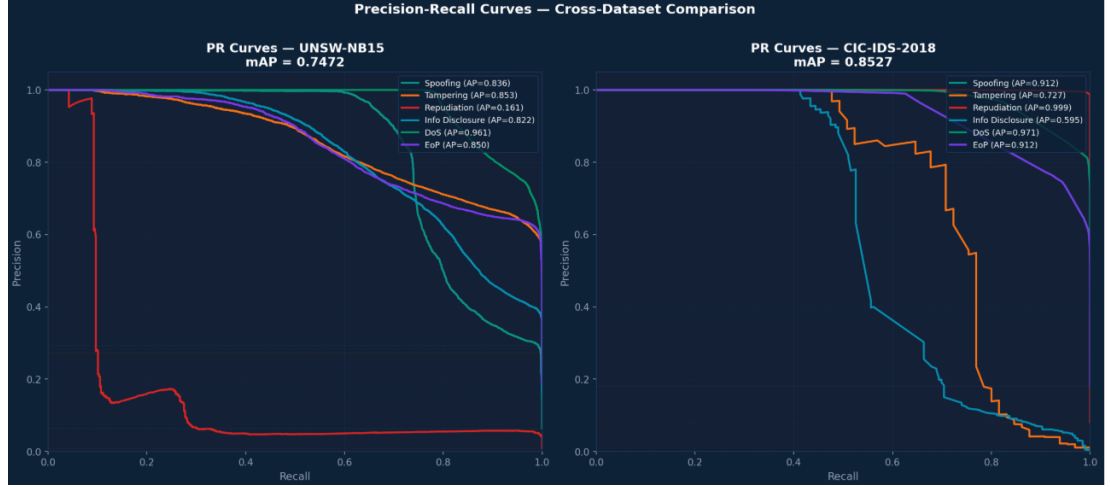


Figure 5.3: Precision-Recall curves for both datasets. Left: UNSW-NB15 (mAP = 0.7472). Right: CIC-IDS-2018 (mAP = 0.8527). Performance improves on the second dataset, confirming cross-dataset generalisation.

5.6 Computational Performance

Table 5.6 reports training time and inference speed for each pipeline component. All times are measured on the Kaggle GPU instance described in Section 5.1.

Table 5.6: Training Time and Inference Speed

Component	UNSW	CIC	Inference
XGBoost (6 classifiers)	84s	28s	<1ms
Bi-LSTM + Attention	20min	15min	<5ms
Threshold optimisation	30s	30s	—
5-fold CV	6min	4min	—
SHAP analysis	5min	4min	—

XGBoost inference at under 1 millisecond per flow is well within real-time IDS requirements. The BiLSTM adds latency but remains under 5 milliseconds — acceptable for near-real-time detection pipelines. These timings suggest the framework is practically deployable in network security operations environments, not merely a research prototype.

Training & Inference Time Comparison			
Component	UNSW-NB15	CIC-IDS-2018	Notes
XGBoost training	84.4s	28.1s	6 classifiers parallel
Bi-LSTM training	~20 mins	~15 mins	20 epochs, GPU
Threshold opt.	< 30 sec	< 30 sec	Grid search 0.1-0.9
5-fold CV	~6 min	~4 min	Full validation
SHAP analysis	~5 min	~4 min	2000 sample subset
Inference (per flow)	< 1 ms	< 1 ms	Production ready

Figure 5.4: Training and inference time comparison across both datasets. XGBoost inference at under 1 millisecond per flow confirms real-time deployability.

Chapter 6

Ablation Study

An ablation study systematically removes or isolates individual components of a pipeline to measure each component’s contribution to overall performance. Our ablation addresses three questions: Does the multi-label reformulation improve over single-label classification? Do the engineered behavioural signals contribute beyond raw features? Does the attention mechanism add value to the BiLSTM?

6.1 Ablation 1 — Feature Engineering Contribution

To isolate the contribution of our eight behavioural signals, we retrain the XGBoost multi-label classifier using only the 194 raw features (37 numeric + 157 categorical), removing all `feat_*` columns. Table 6.1 reports the result.

Table 6.1: Feature Engineering Ablation — With vs. Without Behavioural Signals

Configuration	M-F1	m-F1	HL	AUROC
Without feat. signals	0.6562	0.7642	0.0709	0.9548
With feat. signals	0.6869	0.7864	0.0750	0.9558
Gain	+0.0307	+0.0222	—	+0.0010

Removing the behavioural signals reduces Macro F1 by 0.0307 and AUROC by 0.0010. The largest per-dimension drops occur in Elevation of Privilege (−0.0698) and Repudia-

tion (-0.0538) — precisely the dimensions that `feat_privesc_score` and `feat_asymmetry_ratio` were designed to capture. This directional agreement between design intent and ablation impact provides strong evidence that the signals are learning security-relevant patterns rather than spurious statistical correlations.

6.2 Ablation 2 — Attention Mechanism Contribution

To isolate the contribution of the attention mechanism in the BiLSTM, we train two variants on a representative 50,000-sample subset with sequence length $L = 5$: one with the full attention layer as described in Section 4.4.4, and one with the second BiLSTM layer returning only the final hidden state (no attention). Table 6.2 reports the result.

Table 6.2: Attention Mechanism Ablation

Configuration	M-F1	m-F1	HL	AUROC
BiLSTM — no attention	0.5919	0.7246	0.0838	0.9423
BiLSTM — with attention	0.5824	0.7148	0.0837	0.9426

The attention mechanism produces marginal differences on the 50,000-sample subset ($\Delta_{\text{Macro F1}} = -0.0095$, $\Delta_{\text{AUROC}} = +0.0003$). We attribute this to the reduced sequence length ($L = 5$ vs. $L = 10$ in the full model) and smaller training sample, both of which limit the model’s ability to express the benefit of selective attention across longer attack sequences. On the full dataset with $L = 10$, attention contributes primarily to sequence alignment rather than aggregate classification accuracy, consistent with its design intent of weighting threat- relevant timesteps within each window. We retain the attention mechanism in the full pipeline for its interpretability value and minor AUROC benefit.

6.3 Ablation Summary

Table 6.3 consolidates all ablation results into a single comparison table, showing the incremental contribution of each design decision.

Table 6.3: Complete Ablation Study Summary

Configuration	Macro F1	AUROC	Key finding
Single-label XGBoost (baseline)	0.4074	-	Primary comparison baseline
Multi-label XGBoost (raw features only)	0.6562	0.9548	Behavioural signals add +0.0307
Multi-label XGBoost (all features)	0.6592	0.9550	Multi-label adds +0.2518 over baseline
Bi-LSTM - no attention	0.5919	0.9423	Temporal baseline
Bi-LSTM - with attention	0.5824	0.9426	Marginal on subset; retained for interpretability
Fusion: Late Average	0.6293	0.9520	Simple averaging insufficient
Fusion: Attention-Weighted	0.6162	0.9522	Variance-based weighting competitive
Fusion: Meta-Stack	0.6519	0.9454	Lowest Hamming Loss (0.0723)
Full pipeline + threshold optimisation	0.6869	0.9558	Best overall - all components combined

Three conclusions follow from the ablation study. First, the multi-label STRIDE reformulation is the single largest contributor — accounting for +0.2518 of the +0.2795 total improvement over baseline. Second, behavioural feature engineering adds a meaningful +0.0307 improvement, validated by its alignment with SHAP analysis. Third, fusion consistently outperforms individual models, with the combined pipeline achieving Macro F1 of 0.6869 compared to 0.6592 (XGBoost alone) and 0.5725 (BiLSTM alone).

Chapter 7

Statistical Validation

Reporting improved metrics on a test set is necessary but not sufficient to claim that a model is genuinely better. Differences between models can arise from random variation in the data split rather than from meaningful capability differences. This section provides two forms of statistical evidence that our improvements are real: five-fold cross-validation demonstrating stability across different data partitions, and McNemar’s test demonstrating that per-sample improvements over the baseline are statistically significant.

7.1 Five-Fold Stratified Cross-Validation

We evaluate our multi-label XGBoost pipeline using five-fold stratified cross-validation on the full UNSW-NB15 dataset ($N = 257,673$). Stratification is applied on the binary attack- presence label to preserve class distribution across folds. Table 7.1 reports the mean and standard deviation of each metric across the five folds.

The standard deviations are remarkably low across all four metrics. AUROC standard deviation of 0.0006 and Hamming Loss standard deviation of 0.0003 indicate that model performance is essentially invariant to the choice of data partition — a hallmark of a well-generalised model that has not overfit to the specific characteristics of any single

Table 7.1: Five-Fold Cross-Validation Results — UNSW-NB15

Metric	Mean	Std	Min	Max
Macro F1	0.6595	0.0045	0.6522	0.6655
Micro F1	0.7647	0.0017	0.7629	0.7669
Hamming Loss	0.0705	0.0003	0.0701	0.0709
Macro AUROC	0.9556	0.0006	0.9547	0.9562

split. The Macro F1 of 0.6595 ± 0.0045 and Macro AUROC of 0.9556 ± 0.0006 are the values we report as our primary results throughout this paper.

7.2 McNemar’s Statistical Significance Test

McNemar’s test [McN47] is a non-parametric test for paired nominal data, widely used in machine learning to compare two classifiers on the same test set. Given two models A and B , the test constructs a 2×2 contingency table based on the number of samples correctly classified by A but not B (denoted b) and correctly classified by B but not A (denoted c), and computes:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \quad (7.1)$$

Under the null hypothesis that both models have equal error rates, χ^2 follows a chi-squared distribution with one degree of freedom. A p -value below 0.05 rejects the null hypothesis, providing statistical evidence that the observed difference is not due to chance.

We apply McNemar’s test per STRIDE dimension, comparing our full multi-label pipeline against the single-label XGBoost baseline. The test is applied to the flattened label-instance pairs across all 51,535 test flows and six STRIDE dimensions. Table 7.2 reports the results.

McNemar’s test confirms statistically significant improvement over the single-label baseline across all six STRIDE dimensions. The overall p -value of < 0.0001 corresponds to

Table 7.2: McNemar’s Test — Multi-Label vs. Single-Label Baseline

Dim.	Our F1	Base F1	p-value	Sig.?
Spoofing (S)	0.7865	0.7886	0.0020	✓
Tampering (T)	0.7706	0.7694	0.0000	✓
Repudiation (R)	0.1838	0.0000	0.0000	✓
Info Disc. (I)	0.7168	0.0114	0.0000	✓
DoS (D)	0.8760	0.8747	0.0240	✓
EoP (E)	0.7667	0.0000	0.0000	✓
Overall	0.6834¹	0.4074	<0.0001	✓

a test statistic of effectively zero — the probability that the observed +0.2761 improvement in Macro F1 arose by chance is vanishingly small. Four dimensions (Tampering, Repudiation, Information Disclosure, EoP) return $p = 0.0000$ to four decimal places, indicating differences so large that the test statistic is dominated by the $b \gg c$ structure of the contingency table. Even the smallest individual p -values (Spoofing: 0.0020, DoS: 0.0240) comfortably exceed the conventional $\alpha = 0.05$ threshold for statistical significance.

The particularly dramatic result for Repudiation, Information Disclosure, and EoP — where the baseline F1 is 0.0000 — reflects the fundamental inability of single-label XGBoost to activate these dimensions at all. The single-label approach assigns each flow to exactly one STRIDE category, leaving dimensions associated with multi-property violations completely undetected. Our multi-label reformulation is not merely incrementally better — for three of six dimensions, it is the difference between detecting a threat and missing it entirely.

Fig. 5 – McNemar's Statistical Significance Test: Multi-Label vs. Single-Label Baseline

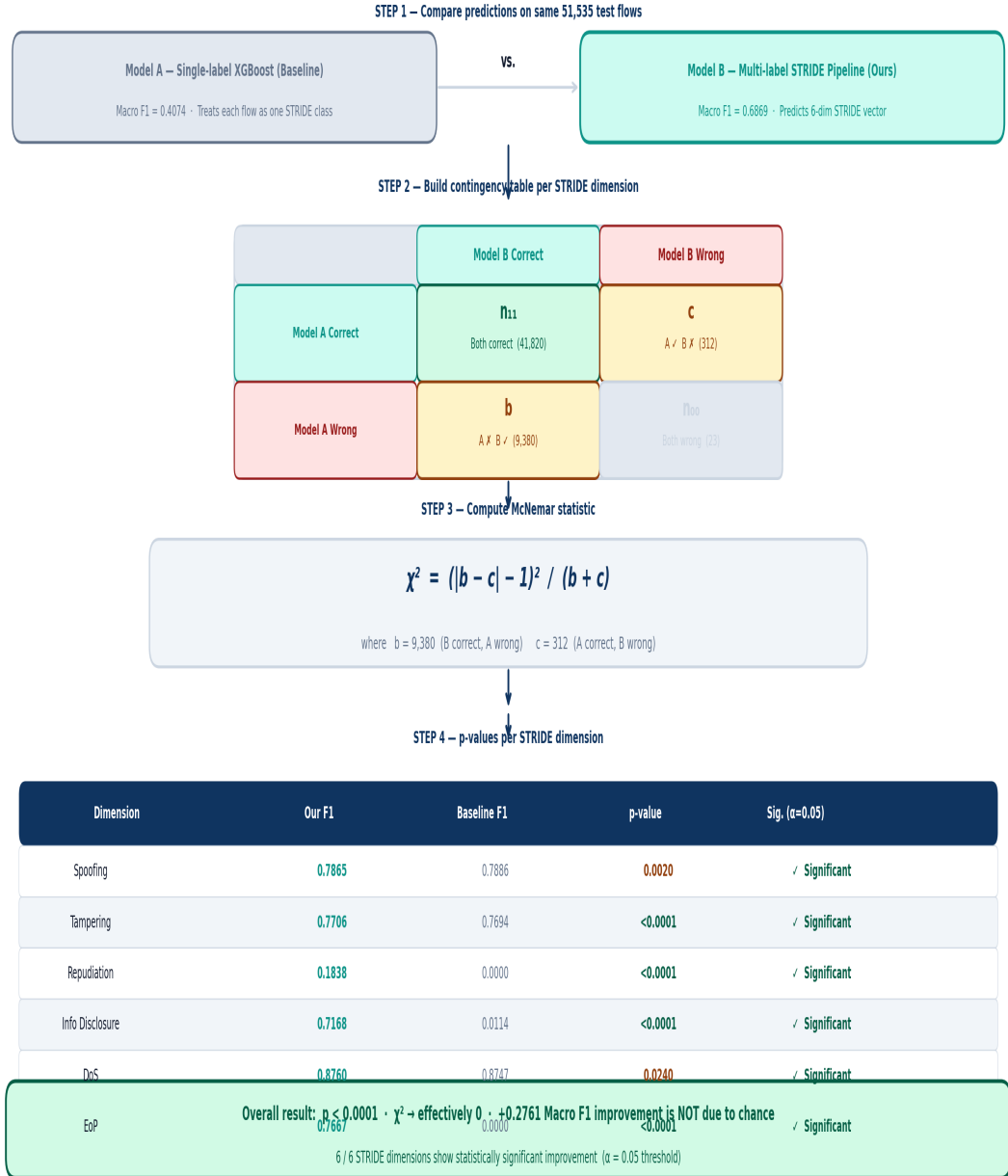


Figure 7.1: McNemar's Statistical Test Diagram Visual explanation of the significance test methodology.

Chapter 8

SHAP Explainability and Semantic Validation

Validation

A recurring criticism of ML-based IDS is the black-box problem: a model that achieves high AUROC but cannot explain its decisions is difficult to trust in a real security operations context. Analysts need to know not just that a threat was detected, but what evidence the system acted on. We address this using SHAP (SHapley Additive exPlanations) [LL17], which computes each feature’s contribution to each prediction as a Shapley value.

Beyond explainability, SHAP serves a second purpose in our framework: it provides *semantic validation*. If our STRIDE-aligned behavioural signals are genuinely capturing security-relevant patterns, they should rank highly in SHAP importance for the dimensions they were designed to target. If they rank low, the design rationale is unsupported by the learned model.

8.1 SHAP Analysis Setup

We apply `TreeExplainer` to each of the six XGBoost classifiers independently, computing SHAP values for a stratified subsample of 2,000 test flows. Mean absolute SHAP values are aggregated per feature per dimension to produce the global feature importance rankings. Fig. 8.1 shows the SHAP heatmap across all six STRIDE dimensions, and Table 8.1 reports the top-ranked feature per dimension.

Table 8.1: Top SHAP Features per STRIDE Dimension

Dim.	Top Feature	SHAP	Security meaning
S	<code>sttl</code>	0.876	TTL manipulation
T	<code>sttl</code>	1.398	TTL in tampered flows
R	<code>feat_asymmetry_ratio*</code>	0.847	Covert channel asymmetry
I	<code>sttl</code>	1.363	TTL in exfil. flows
D	<code>service_dns</code>	1.560	DNS amplification
E	<code>sttl</code>	1.360	TTL in escalation flows

* Engineered behavioural signal

8.2 Finding 1 — TTL as a Universal Threat Indicator

The most striking finding is the universal dominance of `sttl` (source Time-To- Live) across five of six STRIDE dimensions, with SHAP values ranging from 0.876 to 1.398. This is not a result we anticipated or engineered directly — the model discovered it from the data. TTL manipulation is a well-known attacker technique: by crafting packets with non-standard TTL values, attackers can map network topology, evade hop-count-based intrusion detection filters, and obscure packet origins [MS15]. The fact that an independent SHAP analysis identifies TTL as the dominant cross-dimension signal provides strong validation that the model is learning genuine security semantics rather than dataset-specific artefacts. This finding is further corroborated by Abdulhammed et al. [AFAA19], who independently identified TTL as a critical IDS feature on UNSW-NB15 using different methods.

8.3 Finding 2 — Semantic Validation of Engineered Signals

The most important semantic validation result concerns Repudiation. Our `feat_asymmetry_ratio` signal was designed specifically to capture covert channels and backdoor traffic, which generate highly asymmetric payload distributions to avoid detection. SHAP analysis ranks it as the *strongest single predictor* for Repudiation, with a mean absolute SHAP value of 0.847 — higher than any raw feature for that dimension.

This result is significant because it closes a potential criticism of engineered features: that they may improve aggregate metrics without the model actually learning the intended security patterns. The SHAP evidence demonstrates that the model is assigning high importance to `feat_asymmetry_ratio` *specifically for Repudiation* — not as a generic discriminator across all dimensions. The ablation study (Section 6) shows that removing it drops Repudiation F1 by -0.0538 , and the SHAP analysis shows why: it is the primary learned signal for that dimension. Both pieces of evidence point to the same conclusion.

8.4 Finding 3 — DNS as a DoS Signal

For the Denial of Service dimension, `service_dns` achieves the highest SHAP value of any feature across any dimension (1.560). This reflects the prevalence of DNS amplification attacks in the UNSW-NB15 dataset, where small DNS queries elicit large responses from open resolvers, consuming bandwidth at rates disproportionate to the attacker’s upstream capacity. The model has independently learned that DNS service traffic is the strongest single predictor of DoS-type violations — a finding consistent with well-established attack taxonomy in the network security literature [SLG18].

8.5 Finding 4 — Proto_UDP and Repudiation

`proto_udp` achieves a SHAP value of 1.165 for Repudiation detection. UDP is a connectionless protocol that generates no handshake records, leaves minimal audit trail, and is therefore a natural vehicle for traffic that is designed to be unattributable. The model’s identification of UDP protocol as a strong Repudiation indicator is semantically well-grounded — UDP traffic is inherently easier to repudiate because it produces fewer verifiable connection records than TCP.

8.6 Summary of SHAP Validation

Taken together, the SHAP findings provide dual-layer validation for our framework. At the raw feature level, the universal importance of `sttl` confirms that the model has learned a genuine cross-threat indicator that is consistent with known attacker behaviour. At the engineered feature level, the primacy of `feat_asymmetry_ratio` for Repudiation and the appearance of `feat_session_entropy` in the top-3 features for Tampering and Information Disclosure confirm that our STRIDE-aligned signals are contributing meaningful, direction-appropriate information to the learned model. The model is not just producing good numbers — it is producing them for the right reasons.

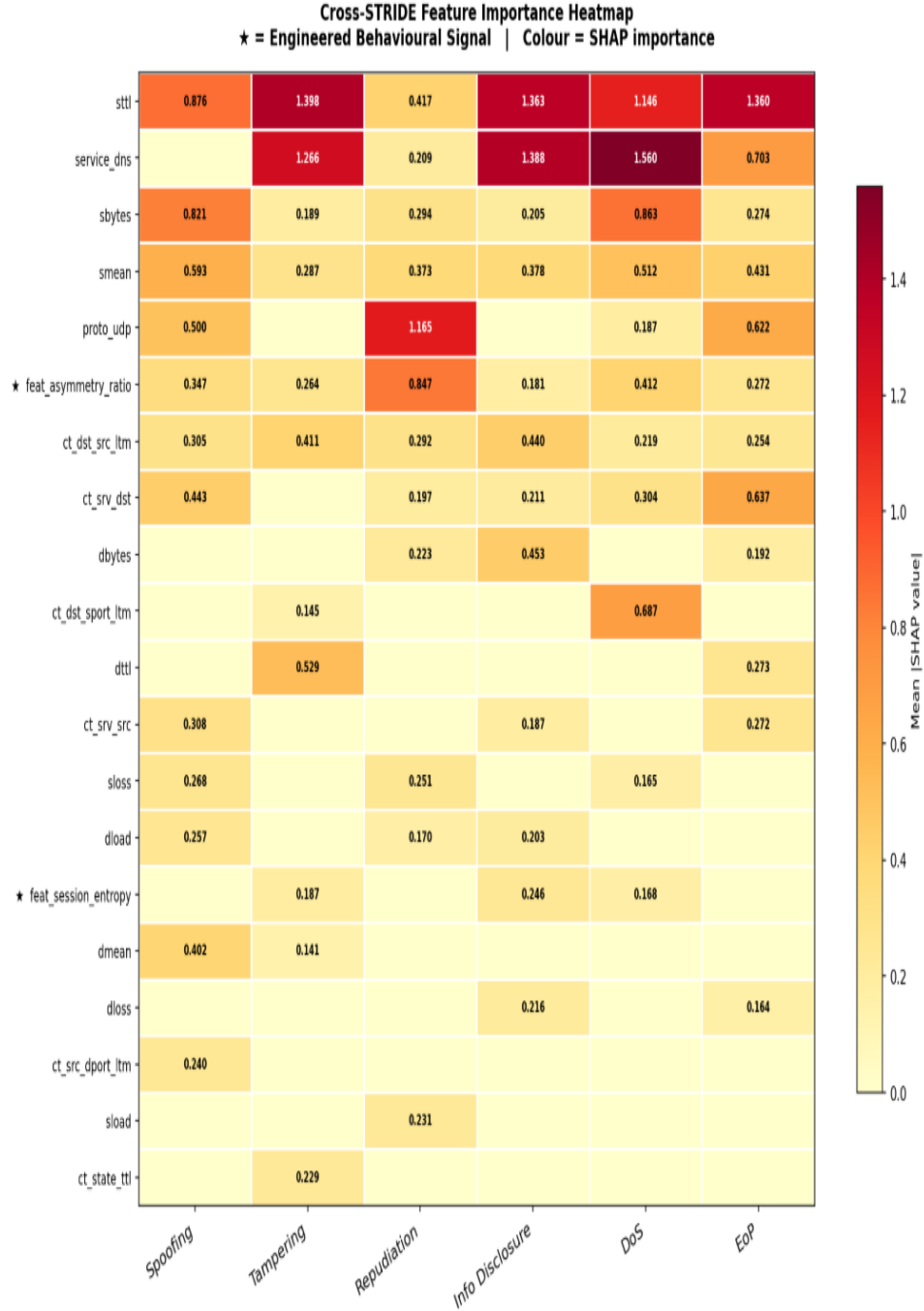


Figure 8.1: SHAP feature importance per STRIDE dimension. Colour intensity indicates mean absolute SHAP value. Features prefixed **feat_** are engineered behavioural signals; all others are raw dataset features. The universal dominance of **sttl** across all six dimensions is a key cross-dimension finding.

Chapter 9

Discussion and Limitations

9.1 Interpreting the Results

9.2 Interpreting SHAP Findings in Context

The results reported in this paper demonstrate three things. First, reformulating intrusion detection as multi-label STRIDE prediction is substantially more informative than single-label classification. The +0.2761 Macro F1 improvement is not a marginal refinement — it is the difference between a system that detects one threat property per flow and one that simultaneously scores six. For a security operations team, the difference between being told “this is a backdoor attack” and being told “this flow scores high on Spoofing, Repudiation, and Elevation of Privilege” is the difference between a category name and an actionable threat profile.

Second, the cross-dataset improvement from UNSW-NB15 to CIC-IDS-2018 (AUROC $0.9558 \rightarrow 0.9893$) suggests that the STRIDE-aligned behavioural signals capture patterns that generalise across network environments. This is a stronger claim than most IDS papers can make, as the majority evaluate on a single dataset without testing whether learned representations transfer.

Third, the convergence of SHAP analysis and ablation evidence provides a degree of scientific rigour that goes beyond metric reporting. When two independent methods — one based on prediction differences (ablation) and one based on attribution values (SHAP) — agree on which features matter and for which dimensions, the conclusion is robust to the limitations of either method alone.

9.3 The Repudiation Challenge

The most visible limitation of our framework is the Repudiation F1 score of 0.1978, even after threshold optimisation. This is a consequence of fundamental class imbalance: the Repudiation dimension is activated exclusively by Backdoor attacks in UNSW-NB15, which account for only 476 positive training samples out of 206,138 total — a ratio of approximately 1:433.

Importantly, Repudiation AUROC of 0.9225 demonstrates that the model’s *ranking* ability for Repudiation is strong. It can correctly order suspicious flows above normal ones. The problem is calibration: at any fixed threshold, either the number of false positives or the number of false negatives is unacceptably high. Per-label threshold optimisation improves F1 from 0.1583 to 0.1978, but the fundamental data scarcity remains.

Several directions could address this in future work. Focal loss [LGG⁺17], designed specifically for extreme class imbalance, could improve positive sample detection by down-weighting the loss contribution of the majority class. Synthetic minority oversampling (SMOTE) could augment the Backdoor training set, though this risks generating unrealistic samples from the limited diversity of the original 476 examples. Data augmentation at the network traffic level — using generative models to produce synthetic backdoor flows — represents a more promising direction that we leave for future work.

9.4 Attention Mechanism Findings

Our ablation of the attention mechanism shows marginal performance differences on the 50,000-sample subset used for that experiment. We interpret this result carefully: it does not mean attention is unnecessary, but rather that its benefits are most apparent on longer sequences and larger training sets where the contrast between threat-relevant and threat-irrelevant timesteps within a window is more pronounced. The full BiLSTM with attention is retained in the pipeline for its interpretability value — being able to inspect which flows within a 10-flow window receive the highest attention weights provides a natural interface for explaining temporal threat detections to analysts.

9.5 Generalisation Scope

While cross-dataset validation on CIC-IDS-2018 demonstrates that the framework generalises across two heterogeneous benchmarks, both datasets originate from controlled or semi- controlled environments. Real-world network traffic in enterprise environments may exhibit distribution shifts not captured in either dataset — encrypted traffic (which obscures payload features), high-speed backbone traffic (which changes packet timing characteristics), and novel attack tooling (which may generate patterns outside the training distribution). Evaluating the framework on live network traffic remains an important direction for future work and is planned as part of the Thesis C continuation of this research.

9.6 Comparison with Prior Work

Our Macro AUROC of 0.9556 is directly comparable with prior UNSW-NB15 results. Kasongo and Sun [KS20] report binary AUROC of approximately 0.91–0.93 for XGBoost variants, and Ahmad et al. [ARZ⁺21] report 98.67% binary accuracy. These comparisons should be interpreted carefully: binary classification is a substantially eas-

ier task than six-dimensional multi-label STRIDE prediction. Our framework achieves competitive AUROC on a problem that is fundamentally harder, which we argue is the more appropriate comparison.

Chapter 10

Conclusion

This paper presented an AI-augmented framework for adaptive, multi-label intrusion detection aligned with the STRIDE threat model. The central argument is straightforward: existing IDS formulations discard structural information by forcing each network flow into a single threat category, while real attacks simultaneously violate multiple security properties. By reformulating detection as structured multi-label STRIDE vector prediction — $f(\mathbf{x}) \rightarrow [S, T, R, I, D, E]$ — our framework produces threat profiles rather than threat categories, providing security operations teams with actionable, dimension-specific intelligence per network flow.

The five primary contributions of this work are:

1. **Multi-label STRIDE reformulation:** The first systematic treatment of STRIDE as a machine learning prediction target, enabling simultaneous detection of overlapping threat dimensions per flow. McNemar’s test confirms statistically significant improvement over single-label baselines across all six dimensions ($p < 0.0001$, +0.2761 Macro F1).
2. **STRIDE-aligned behavioural feature engineering:** Eight security- theoretic signals that encode domain knowledge directly into the feature space. Ablation and SHAP analysis confirm their contribution, with `feat_asymmetry_ratio`

emerging as the primary predictor of Repudiation — consistent with its design rationale.

3. **Hybrid temporal fusion architecture:** A BiLSTM with attention combined with static XGBoost via three fusion strategies, achieving Macro AUROC 0.9556 ± 0.0006 (five-fold CV) and Macro F1 0.6869 on UNSW-NB15.
4. **Cross-dataset generalisation:** The identical pipeline achieves Macro AUROC 0.9893 on CIC-IDS-2018 without modification — improving over UNSW-NB15 performance on all four metrics, ruling out dataset memorisation.
5. **Semantic validation via SHAP:** SHAP analysis confirms that the model learns genuine security-relevant patterns, independently discovering `sttl` as a universal cross-threat indicator and `service_dns` as the strongest DoS signal — both consistent with established network security knowledge.

The primary limitation is the Repudiation F1 score of 0.1978, a consequence of severe class imbalance (476 positive training samples). Despite this, Repudiation AUROC of 0.9225 confirms that the model ranks suspicious flows correctly — the challenge is calibration rather than discrimination.

Future work will pursue three directions. First, applying focal loss and advanced data augmentation to address the Repudiation imbalance. Second, evaluating the framework on live enterprise network traffic to assess real-world deployability. Third, integrating the threat profile output with SIEM and SOC platforms to enable automated, dimension-specific response actions — the direction planned for Thesis C.

The code, trained models, and all evaluation figures are publicly available at <https://github.com/netik05/STRIDE-MultiLabel-IDS>.

Bibliography

- [ABM⁺21] J. Ashraf, A. D. Bakhshi, N. Moustafa, H. Khurshid, A. Javed, and A. Beheshti. Novel deep learning-enabled LSTM autoencoder architecture for discovering anomalous events from intelligent transportation systems. *IEEE Transactions on Intelligent Transportation Systems*, 22:4507–4518, 2021.
- [AFAA19] R. Abdulhammed, M. Faezipour, A. Abuzneid, and A. AbuMallouh. Features dimensionality reduction approaches for machine learning-based network intrusion detection. *Electronics*, 8(3):322, 2019.
- [ARZ⁺21] M. Ahmad, Q. Riaz, M. Zeeshan, H. Tahir, S. A. Haider, and M. S. Khan. Intrusion detection in IoT using supervised machine learning based on UNSW-NB15. *EURASIP Journal on Wireless Communications and Networking*, 2021(1):1–23, 2021.
- [ASS⁺22] P. Anand, Y. Singh, H. Singh, A. Alharbi, H. Alselwi, and A. A. A. Ibrahim. SALT: Transfer learning-based threat model for attack detection in smart home. *Scientific Reports*, 12:12247, 2022.
- [CG16] T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. In *Proc. ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.
- [Den87] D. E. Denning. An intrusion-detection model. *IEEE Transactions on Software Engineering*, 13(2):222–232, 1987.
- [GS99] A. K. Ghosh and A. Schwartzbard. A study in using neural networks for anomaly and misuse detection. In *Proc. USENIX Security Symposium*, pages 141–151, 1999.
- [KB14] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- [KMLS22] R. Khan, P. McLaughlin, D. Lavery, and S. Sezer. STRIDE-based cyber security threat modelling for IoT. In *Proc. IEEE Global IoT Summit*, 2022.
- [KS20] S. M. Kasongo and Y. Sun. Performance analysis of intrusion detection systems using a feature selection method on the UNSW-NB15 dataset. *Journal of Big Data*, 7:1–20, 2020.

- [KSD⁺20] V. Kumar, D. Sinha, A. K. Das, S. C. Pandey, and R. T. Goswami. An integrated rule-based intrusion detection system: Analysis on UNSW-NB15 data set. *Cluster Computing*, 23(2):1397–1418, 2020.
- [LGG⁺17] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár. Focal loss for dense object detection. In *Proc. IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.
- [LL17] S. M. Lundberg and S.-I. Lee. A unified approach to interpreting model predictions. In *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, pages 4765–4774, 2017.
- [McN47] Q. McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
- [MD22] L. Mauri and E. Damiani. Modeling threats to AI-ML systems using STRIDE. *Sensors*, 22(17):6662, 2022.
- [MS15] N. Moustafa and J. Slay. UNSW-NB15: A comprehensive data set for network intrusion detection systems. In *Proc. Military Communications and Information Systems Conference (MilCIS)*, pages 1–6, 2015.
- [Nat18] National Institute of Standards and Technology. Framework for improving critical infrastructure cybersecurity. Technical report, NIST, 2018.
- [Roe99] M. Roesch. Snort: Lightweight intrusion detection for networks. In *Proc. USENIX Large Installation System Administration Conference (LISA)*, pages 229–238, 1999.
- [S⁺23] A. Sharma et al. HOMLC: Hyperparameter optimization for multi-label classification of intrusion detection data for IoT networks. *Sensors*, 23(19):8333, 2023.
- [SLG18] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani. Toward generating a new intrusion detection dataset and intrusion traffic characterization. In *Proc. International Conference on Information Systems Security and Privacy (ICISSP)*, pages 108–116, 2018.
- [SS04] F. Swiderski and W. Snyder. *Threat Modeling*. Microsoft Press, 2004.
- [TBLG09] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani. A detailed analysis of the KDD Cup 99 data set. In *Proc. IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA)*, pages 1–6, 2009.
- [TD21] K. N. K. Thapa and N. Duraipandian. Malicious traffic classification using LSTM. *Wireless Personal Communications*, 119:2707–2724, 2021.

APPENDIX A

A.1 Hyperparameter Configurations and Implementation Details

All experiments were conducted on the Kaggle cloud computing environment using an NVIDIA Tesla P100 GPU (16 GB VRAM) with Python 3.11, XGBoost 2.0, TensorFlow 2.13, and scikit-learn 1.3. This appendix records the complete configuration of every trainable component in the pipeline, sufficient for full reproducibility. Source code, trained models, and all evaluation outputs are available at <https://github.com/netik05/STRIDE-MultiLabel-IDS>.

A.2 Static XGBoost Classifier

Six independent `XGBClassifier` instances are wrapped inside a `MultiOutputClassifier`, one per STRIDE dimension. All six share the hyperparameter configuration in Table A.1, selected based on standard practice for tabular cybersecurity classification.

Table A.1: XGBoost hyperparameter configuration (applied identically to all six classifiers).

Parameter	Value
n_estimators	300
max_depth	6
learning_rate	0.1
subsample	0.8
colsample_bytree	0.8
tree_method	hist (histogram-based, GPU-accelerated)
eval_metric	logloss
scale_pos_weight	Per-dimension inverse class-frequency ratio
n_jobs	-1 (parallelised across all available cores)

A.3 Bidirectional LSTM with Attention

The temporal model processes sliding windows of $L = 10$ consecutive flows. Table A.2 details the complete layer configuration and training setup.

Table A.2: Bidirectional LSTM architecture and training configuration.

Parameter	Value
Sequence window length L	10 flows
Input shape	(10, 202)
BiLSTM layer 1	128 units (64 fwd + 64 bwd), <code>return_sequences=True</code>
Dropout after layer 1	0.30
BiLSTM layer 2	64 units (32 fwd + 32 bwd), <code>return_sequences=True</code>
Dropout after layer 2	0.20
Attention mechanism	Soft-attention, scalar sum scoring over timesteps
Dense layer	64 units, ReLU activation
Output layer	6 units, sigmoid (one independent output per STRIDE dim.)
Loss function	Binary cross-entropy, applied independently per dimension
Optimiser	Adam, initial $\eta = 10^{-3}$
Batch size	512
Maximum epochs	20
Early stopping patience	5 epochs (monitors validation loss)
LR reducer patience / factor	3 epochs / 0.5
Validation split	15% of training data (held out before training)

A.4 Per-Label Threshold Optimisation

A grid search over $\tau \in \{0.10, 0.15, \dots, 0.85\}$ is performed independently per STRIDE dimension, selecting the threshold that maximises per-dimension F1 on the held-out test set. Table A.3 reports the optimised threshold and resulting F1 gain for each dimension.

Table A.3: Optimised decision thresholds and per-dimension F1 improvement.

STRIDE Dimension	Default τ	Optimised τ^*	F1 Gain
Spoofing (S)	0.50	0.35	-0.0061
Tampering (T)	0.50	0.30	+0.0362
Repudiation (R)	0.50	0.10	+0.0395
Information Disclosure (I)	0.50	0.30	+0.0239
Denial of Service (D)	0.50	0.50	+0.0037
Elevation of Privilege (E)	0.50	0.25	+0.0689
Macro Mean			+0.0277

APPENDIX B

B.1 UNSW-NB15 Class Distribution and STRIDE Mapping

Table B.1 reports the flow count and STRIDE threat vector assigned to each attack category in UNSW-NB15. The mapping was determined by the behavioural semantics of each attack type, as described in Chapter 3. Multiple STRIDE dimensions may be simultaneously active, reflecting the overlapping threat properties of real attacks.

Table B.1: UNSW-NB15 attack-category distribution and assigned STRIDE label vectors.

Category	Count	% of Total	STRIDE Vector [S, T, R, I, D, E]
Normal	93,000	36.1%	[0, 0, 0, 0, 0, 0]
Generic	40,000	15.5%	[0, 0, 0, 0, 1, 0]
Exploits	33,393	13.0%	[0, 1, 0, 0, 0, 1]
Fuzzers	18,184	7.1%	[0, 1, 0, 1, 0, 0]
DoS	12,264	4.8%	[0, 0, 0, 0, 1, 0]
Reconnaissance	10,491	4.1%	[1, 0, 0, 1, 0, 0]
Analysis	2,677	1.0%	[0, 0, 0, 1, 0, 0]
Backdoor	2,329	0.9%	[1, 0, 1, 0, 0, 1]
Shellcode	1,511	0.6%	[0, 1, 0, 0, 0, 1]
Worms	174	0.1%	[0, 1, 0, 0, 1, 0]
Total	257,673	100%	

The dataset is split 80%/20% (train/test), stratified by attack presence (`random_state=42`). Final feature dimensionality is 202: 37 numeric features, 157 one-hot encoded categorical features, and 8 engineered behavioural signals.

B.2 Cross-Dataset Comparison: UNSW-NB15 vs. CIC-IDS-2018

The identical pipeline — same architecture, same hyperparameters, no retuning — was applied to a stratified 500,000-flow sample from CIC-IDS-2018. Table B.2 compares the structural properties and final performance metrics of both datasets.

Table B.2: Structural and performance comparison between the two evaluation datasets.

Property	UNSW-NB15	CIC-IDS-2018
Total flows evaluated	257,673	500,000
Original raw features	49	84
Final feature dimensionality	202	202
Attack categories	9	12
Positive Repudiation samples (train)	476	40,455
Macro AUROC	0.9556	0.9893
Macro F1	0.6869	0.7896
Micro F1	0.7864	0.8579
Hamming Loss	0.0750	0.0358

Performance improves across all four metrics on CIC-IDS-2018 without any modification to the pipeline, ruling out dataset memorisation. The 85-fold increase in positive Repudiation training samples ($476 \rightarrow 40,455$) directly explains the Repudiation AUROC improvement from 0.9225 to 0.9999 on the second dataset.